

Taxing Wealth Unlocks Capital Gains: An Efficiency Rationale for Wealth Taxes^{*}

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Abstract

Realization-based capital income taxation generates *capital lock-in* because investors have an incentive to delay tax payments. This generates inefficiency as investors forgo investment opportunities that require them to realize capital gains. We show that *wealth taxation can unlock capital*, eliminating the distortion on investors' portfolio choice. Moreover, wealth taxes do not distort this choice in the absence of capital income taxation. This provides an efficiency rationale for wealth taxation independent of equity: shifting revenues from capital income to wealth taxes reduces distortions on portfolio choice. We provide conditions for the optimal tax mix between capital income and wealth taxes that balance the equity gains from both taxes against efficiency losses related to savings and portfolio choices. These conditions hinge on the *cross-base elasticity* of capital income to wealth taxes which captures lock-in responses.

JEL classification: H24, D14, G51, H21, M21.

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1. Introduction

Capital gains are usually taxed on realization rather than accrual, and dividends only when shareholders elect payouts. Such realization-based taxes imply that investors have substantial control over the timing of tax payments. They also give investors an incentive to delay the realization of capital gains to avoid taxation.¹ Capital income taxes therefore generate *capital lock-in*: investors delay realizing gains to reduce the present value of tax obligations, even when doing so means missing out on profitable investments (see, for example, [Auerbach 1991](#)).

In this context, we provide a new rationale for wealth taxation based on the optimality of investors' portfolio choices, independent of society's views on inequality. A wealth tax can unlock capital locked in by capital income taxes while leaving portfolio choices unaffected in the absence of capital income taxes. The key mechanism is simple. The wealth tax makes paying capital income taxes today more attractive by making the *present value* of tax obligations independent of the realization choice, counteracting the incentive to delay gains from capital income taxation. In this way, shifting revenue from capital income tax to wealth taxes can raise efficiency.

We formalize this result in a model with heterogeneous investors that differ along three dimensions: (i) the principal of their initial investment, (ii) their initial capital gain, and (iii) the return from keeping their initial investment to maturity. Investors can borrow or lend freely at the market interest rate. Hence, they choose between keeping the initial investment and selling it to lend.² Interest income and realized capital gains

¹There is strong evidence that retained earnings are causally connected to realization-based taxes. Using US state-level data, [Agersnap and Zidar \(2021\)](#) find a negative relationship between the capital-gains tax rate and realized capital gains. For the 2013 tax hike, [Saez \(2017\)](#) shows that the reduction in realized capital gains is primarily driven by a timing effect. In Greece and France respectively, [Bilicka, Güçeri, and Koumanakos \(2025\)](#); [Bach, Bozio, Fabre, Guillouzouic, Leroy, and Malgouyres \(2024\)](#) find that following a dividend tax hike, closely-held firms reduce dividend payments, and reinvest in financial assets substituting for portfolio savings at the private level.

²Our results hold more broadly, as we show in Appendix E where we extend our setup to include stochastic returns and partial realization of capital gains.

are taxed at the same rate. We refer to this tax as the capital income tax.

Efficient portfolio choice requires investors to sell whenever their investment's return lies below the market-interest rate. The capital income tax distorts this decision because postponing the sale reduces the net-present value of tax payments on the initial capital gain. Hence, the tax induces some investors to hold low-return assets. This portfolio inefficiency captures capital lock-in.

The wealth tax reduces this distortion by lowering investors' effective discount rate. When the wealth tax rate equals $\tau_a^* \equiv (1 - \tau_k) r$ —where τ_k is the capital income tax rate and r is the market-interest rate—investors do not discount future tax payments. Thus, wealth taxes take away the incentive for deferral as the present value of the tax bill becomes independent of the portfolio choice. This result reflects the fact that the wealth tax bill in the second period rises one-for-one with taxes deferred from the first period, so the decrease in capital income taxes from delaying realization is canceled out by additional wealth tax obligations.³ In this way, setting the wealth tax rate at τ_a^* restores efficient portfolio choice: investors realize their capital gains if their own rate of return is lower than the market rate. For an interest rate of 2.5 percent, and a capital income tax rate of 35 percent, a wealth tax of 1.6 percent eliminates portfolio distortions.

In the second part of the paper, we extend the analysis to optimal taxation. Optimal taxes balance societal preferences for redistribution with the distortions introduced by taxes. Namely, the realization-based income tax creates a portfolio distortion that wealth taxation mitigates. However, both the income tax and the wealth tax discourage savings. The presence of a portfolio distortion makes *cross-base elasticities* play a central role in determining optimal taxes. That is, how much does the capital income tax base respond to a change in the wealth tax rate.

We derive a simple sufficient condition under which it is always optimal to set

³This mechanism is different from proposals to tax capital income on accrual and instead is based on the intertemporal interactions between taxes. We discuss accrual-based taxes in Section 5.

a positive wealth tax: if the *cross-elasticity* of taxable capital income with respect to the wealth tax is negative, the efficiency gain from mitigating lock-in outweighs the intertemporal distortion of the wealth tax. Crucially, this differs from the *own-elasticity* of taxable wealth commonly estimated in the literature on the effects of wealth taxes (see, among others, [Seim 2017](#); [Jakobsen, Jakobsen, Kleven, and Zucman 2020](#); [Londoño-Vélez and Ávila-Mahecha 2021](#); [Ring 2024](#)). The distortion on the wealth tax base is negligible when the wealth tax rate is close to zero, but the cross-base elasticity remains first order as the wealth tax interacts with the realization decision of investors.

Finally, we discuss how wealth-tax design shapes our efficiency result. Common proposals and implementations tax the *market value of wealth*.⁴ This conflates a tax on the historical value of wealth (the value of the principal) and a tax on accrued capital gains, creating double taxation when gains are eventually realized. We show that even replacing this component with a tax on unrealized capital gains is costly as it generates capital *lock-out*, undoing *lock-in* at the cost of further distortions on portfolio choice.

By contrast, a tax on the historical value of wealth does not distort portfolio choice on its own. It generates no capital lock-in or lock-out in the absence of capital income taxation. It also avoids real-time capital-gain valuation, simplifying its implementation relative to market-value wealth taxes or accrual-based capital income taxes.

Related literature. Our analysis provides a new perspective on how the design of capital taxation affects the economy that complements an extensive literature on capital income and wealth taxation (see [Scheuer and Slemrod 2021](#) for a recent survey). While our results are based around the efficiency consequences of capital income and wealth taxes, investors' control over the timing of tax payments also has a profound effect on the progressiveness of the tax system. Using US tax returns and data from the Fortune

⁴See [Saez and Zucman \(2019\)](#) for a formal wealth tax proposal and [California Secretary of State \(2025\)](#); [U.S. Congress, Senate \(2026\)](#) for recent policy proposals. Wealth taxes on the market value of wealth are currently in effect in Colombia, Norway, Spain, and Switzerland among others.

400, [Yagan \(2023\)](#) and [Balkir, Saez, Yagan, and Zucman \(2025\)](#) find that the effective income tax rate of the 400 wealthiest American households lies significantly below that of the average American when accounting for unrealized income. Similarly, research from France ([Bach, Bozio, Fabre, Guillouzouic, and Malgouyres 2025](#)), Italy ([Guzzardi, Palagi, Roventini, and Santoro 2024](#)), Norway ([Aaberge, Francesconi, Modalsli, and Vestad 2024](#)), and the Netherlands ([Bruil, Van Essen, Leenders, Lejour, Möhlmann, and Rabaté 2022](#)), find tax regressivity towards the top of the income distribution. The sharp difference between statutory and effective tax rates largely reflects control of retained earnings and unrealized capital gains at the top of the distribution. As a result, [Slemrod and Chen \(2023\)](#) refer to the capital gains tax as the *Achilles heel* of taxing the rich.

Three related papers that also highlight an efficiency motivation for levying a wealth tax are [Guvenen et al. \(2023, 2024\)](#), who develop models with heterogeneous returns and liquidity constraints, and [Coven, Golder, Gupta, and Ndiaye \(2025\)](#), who study property markets with levered homeowners. The first two papers show that a wealth tax can improve efficiency by reallocating funds from low- to high-productivity entrepreneurs and by incentivizing innovation and managerial effort.⁵ The third paper applies these ideas to housing markets where capital income taxes generate lock-in and shows empirically and quantitatively that property taxes alleviate it, reallocating housing from older to younger households.

Our contribution is complementary: we abstract from liquidity constraints and instead focus on portfolio inefficiency from realization-based capital-gains taxation. We show a wealth tax also enhances efficiency by mitigating capital lock-in. As with the book-value wealth tax in [Guvenen et al. \(2023, 2024\)](#), the key to the efficiency properties of the historical value wealth tax we study is that it does not tax contemporaneous returns. Hence, these papers jointly demonstrate that an appropriately designed wealth

⁵Some of these ideas, such as the use-it-or-lose-it effect of wealth taxes, are mentioned in [Allais \(1977\)](#) and [Piketty \(2013\)](#) but their implications are not studied.

tax can improve efficiency in capital markets.

Our paper also relates to alternative approaches to reducing lock-in, such as taxation of Haig–Simons income, retrospective taxation, or withholding schemes (e.g. [Auerbach 1991](#); [Saez, Yagan, and Zucman 2021](#)). Although these proposals have attractive theoretical properties, they have rarely been implemented in practice. In contrast, wealth taxes have been levied in several developed and developing countries, providing a feasible and well-tested policy instrument. Beyond their redistributive role, highlighted by [Saez and Zucman \(2019\)](#); [Piketty, Saez, and Zucman \(2023\)](#); [Zucman \(2024\)](#), our analysis highlights that wealth taxes can also restore efficiency by mitigating lock-in distortions without relying on taxation based on the accrual of capital gains.

An alternative to the approaches above is put forth by [Aguiar, Moll, and Scheuer \(2024\)](#), who propose a realization-based tax on *net trades*. Under this tax, capital gains are *realized* only when sales are not offset by new asset purchases and depend only on sale prices. This elegant redesign of the capital income tax also eliminates lock-in and has attractive theoretical properties.

Our paper operates on a different reform margin entirely. Rather than redesigning the capital income tax, we ask what can be done to restore efficiency within the existing fiscal architecture. Our approach relies on an instrument already in operation in multiple countries, requiring no new information about investors' asset purchase prices or trading positions, and delivers applicable design recommendations (for example, taxing historical versus market value). The two papers are therefore complementary: [Aguiar, Moll, and Scheuer \(2024\)](#) characterize the optimal structure of the capital income tax, while we characterize the optimal design of a wealth tax that works alongside the existing capital income tax. Together, they illustrate that lock-in can be addressed through reforms at different points in the tax system.

Return heterogeneity is central to our analysis of capital lock-in. Its relevance is well established: return differences are large and persistent across the wealth distribution

(Daminato and Pistaferri 2024; Fagereng et al. 2020; Bach, Calvet, and Sodini 2020), particularly among owners of private firms (Smith, Yagan, Zidar, and Zwick 2019; Becker and Hvide 2022; Smith, Zidar, and Zwick 2023; Hubmer, Halvorsen, Salgado, and Ozkan 2024). Moreover, return heterogeneity is key to explaining wealth dynamics (Atkeson and Irie 2022; Stachurski and Toda 2019; Benhabib and Bisin 2018; Gabaix et al. 2016) and it changes the efficiency implications of capital taxation (Guvenen et al. 2023, 2024).

Accordingly, our analysis also connects to a growing theoretical literature on optimal capital taxation under return heterogeneity. Gerritsen, Jacobs, Spiritus, and Rusu (2025) characterize the optimal taxation of capital income when rates of return differ systematically across investors, while Ferey, Lockwood, and Taubinsky (2024) develop sufficient-statistics formulas for optimal nonlinear taxes under general across-individual heterogeneity. In contrast to these studies, we ask how return heterogeneity shapes the efficiency properties of existing realization-based systems.

Finally, our work relates to a longstanding literature on realization-based taxation. In our model, we impose restrictions on the capital market by not allowing investors to hedge their position, for instance through short-sales or derivatives. Hedging allows investors to close the exposure to their investment without realizing their tax position. As such, they can forego paying realization-based taxation altogether (Constantinides 1983; Stiglitz 1983). Our setup is therefore closer to Auerbach (1991), in which realization is forced in the final period. However, we extend his framework by introducing investor heterogeneity, wealth taxation, and optimal taxation.

2. Set-up

We consider a two-period model with time indexed by $t \in \{1, 2\}$ and a unit mass of investors indexed by i . Investors derive utility from their consumption in every period, C_t^i , with their preferences represented by a utility function $U(C_1^i, C_2^i)$ that is

twice differentiable, increasing, and concave in both arguments. We further assume that consumption in both periods is a normal good.

Investors differ in their wealth endowment A^i (equivalently, the principal or basis for their assets) and in their returns on their wealth in each period, R_t^i . Returns are realized by selling the assets and thus capture the change in their price or market value. The market value is then $W_1^i \equiv (1 + R_1^i) A^i$; which is the sum of the asset's historical price at the time of purchase, A^i , and capital gains $R_1^i A^i$.⁶ Differences in returns reflect differences in the types of assets investors own and their profitability; most notably this includes returns from closely-held corporations fully controlled by the investor.

Investors can choose not to realize their period one returns or to sell a fraction ϕ^i of their assets and realize part of them. Retaining returns can be to the investors' advantage depending on their period two returns and the way capital gains are taxed. Additionally, investors can borrow or lend freely via a risk-free asset, B^i , with market return r . Borrowing, setting $B^i < 0$, allows them to finance consumption in the first period without having to realize their returns or to invest at market returns (which is optimal when R_2^i is low). The choice of ϕ^i and B^i captures the investors' savings and portfolio choices. The domain for returns includes the market return, $R^{\min} < r < R^{\max}$. So, some investors have returns above the market rate, $R_2^i > r$, and some below, $R_2^i < r$.

Our two-period setting with deterministic returns facilitates the exposition and makes our results more transparent. However, none of our key results rely on this assumption, as we show in Appendix E where we extend our setting to a finite horizon T and introduce uncertainty by making investors face a stochastic return process with a general time-dependent drift.

⁶Changes in returns or prices can reflect real dividends and fluctuations in pricing driven, for instance, by discount factors. We assume throughout that taxes do not directly change these returns (or asset prices). This is the case in production economies like that in Guvenen et al. (2024) and more generally in arbitrage-free financial markets (Ross 1987; Bjerk Sund and Schjelderup 2022, 2026).

2.1. Tax instruments

The investors face four policy instruments. The first one is a tax on *capital income* at rate τ_k . Capital income is composed of realized capital gains and interest income net of interest expenditures.⁷ The second one is a tax on *wealth* at rate τ_a . This tax falls on the historical value of wealth (or on the principal of investments in each period), A^i in the first period and $(1 - \phi^i) (1 + R_1^i) A^i + B^i$ in the second. The third one is a tax on *unrealized capital gains* at a rate τ_R . This tax falls on the returns generated by the investors that are not cashed out, $(1 - \phi^i) R_1^i A^i$. Finally, there is a *lump-sum transfer* (or tax) T that the investor receives (pays) in period 1.

Two of the instruments warrant some more discussion. First, the wealth tax we study is a tax on the historical value of wealth, A^i , rather than on its market value, W^i .⁸ This distinction is relevant for investors' decisions. As highlighted in [Guvenen et al. \(2023\)](#) and [Guvenen et al. \(2024\)](#), taxing the market value of wealth conflates the effects of wealth taxes with those of taxes on capital gains because market values depend on current and future returns. We return to these issues in Section 5.

Second, we introduce taxes on unrealized capital gains, τ_R , because of our focus on the lock-in effect generated by capital taxes. While this tax is impractical (chiefly due to implementation issues) and often suboptimal (see, for example, [Aguiar, Moll, and Scheuer 2024](#)), it turns out to be instructive about the incentives for lock-in as it directly targets capital gains. Considering a tax on unrealized capital gains also lets us clarify the way in which wealth taxes *unlock* them, as we show in Section 3.

⁷Dividends from closely-held companies are part of capital income and taxed at the same rate. This form of taxation also implies that capital gains are only taxed if they are realized into income. For example, “*paper gains*” where prices go up in period 1 and down in period 2 would not be taxed by τ_k if the investor does not realize the period 1 gains by selling (part of) their asset. However, doing so also lowers overall income. Wealth taxes disincentivize this behavior.

⁸The Norwegian wealth tax is based on historical value, much like the wealth tax we study ([Norwegian Ministry of Finance 2020](#)).

2.2. Portfolio choice and the intertemporal budget constraint

The investors' portfolio choice determines their resources in the first period. Investors choose their borrowing and the fraction of their assets to sell, realizing a portion of their capital gains. These choices determine their access to liquidity to fund consumption and tax expenditures. Their period 1 budget constraint is

$$\phi^i (1 + R_1^i) A^i - B^i + T = C_1^i + \tau_k \overbrace{\phi^i R_1^i A^i}^{\text{Realized Cap. Gains}} + \tau_a A^i + \tau_R \underbrace{(1 - \phi^i) R_1^i A^i}_{\text{Unrealized Cap. Gains}}. \quad (1)$$

Choosing to realize more of their capital gains (higher ϕ^i) gives investors more funds (first term in the left-hand side) but also represents costs from taxes on capital gains (second term in the right-hand side). Taxes on unrealized capital gains (last term in the right-hand side) provide an incentive for the realization of gains.

The portfolio choice of the first period also determines the returns in the second period. Investors accrue a capital gain on the unsold fraction of their initial assets and they earn interest income from the risk-free asset (or pay it if they borrowed). In period 2, investors sell off their remaining assets and use the proceeds to finance consumption and tax expenditure. Accordingly, all capital gains are realized in the final period as in [Auerbach \(1991\)](#).⁹ An important implication of this is that unrealized capital gains taxes do not affect returns in the second period and hence have no effect on investors' savings choices.

⁹The terminal-realization assumption works in favor of realization-based taxation. In infinite-horizon models of realization-based taxation, such as [Constantinides \(1983\)](#) and [Stiglitz \(1983\)](#), investors may continue to defer gains rather than trigger the tax, so the lock-in margin includes the possibility of avoiding realization for arbitrarily long horizons. Forcing realization in period 2 guarantees that deferred gains are eventually taxed, limiting the extent to which realization-based taxation can be avoided.

Wealth in period 2, before taxes, is

$$W_2^i \equiv (1 + R_2^i) (1 - \phi^i) (1 + R_1^i) A^i + (1 + r) B^i. \quad (2)$$

Capital gains are computed with respect to the initial value of the (unsold) assets and the principal of the risk-free asset (either bought or sold), that is, $W_2^i - (1 - \phi^i) A^i - B^i$. Wealth taxes fall on the value of unsold wealth in period 1, $(1 - \phi^i) (1 + R_1^i) A^i + B^i$, that is, on the principal for period 2 returns. The second period budget is

$$(1 + \tilde{R}_2^i) \underbrace{(1 + R_1^i) (1 - \phi^i) A^i + (1 + \tilde{r}) B^i}_{\text{Value of Unsold Assets}} = C_2^i + \underbrace{\tau_k R_1^i (1 - \phi^i) A^i}_{\text{Deferred Cap. Income}}, \quad (3)$$

where $\tilde{R}_2^i \equiv (1 - \tau_k) R_2^i - \tau_a$ denotes the effective return on the unsold assets after taxes are paid and $\tilde{r} \equiv (1 - \tau_k) r - \tau_a$ the effective return on the risk-free asset. Finally, the second term on the right-hand side is the deferred income tax on the portion of the capital gain that remained unrealized in period 1.

The inter-temporal budget constraint is

$$\underbrace{\left(\phi^i + \frac{1 + \tilde{R}_2^i}{1 + \tilde{r}} (1 - \phi^i) \right) (1 + R_1^i) A^i + T}_{\text{NPV Assets}} = C_1^i + \frac{C_2^i}{1 + \tilde{r}} + \tau_a A^i + \underbrace{\left(\left(\tau_R + \frac{\tau_k}{1 + \tilde{r}} \right) (1 - \phi^i) + \tau_k \phi^i \right) R_1^i A^i}_{\text{NPV tax on first period return}}. \quad (4)$$

The net present value of assets comes from a fraction ϕ^i of initial assets realized in period 1 and the discounted second-period returns of the remaining fraction. The net present value of taxes on the first period returns, R_1^i , has three components: the tax on realized gains τ_k and unrealized gains τ_R in period 1, and the tax on realized gains in period 2, $\frac{\tau_k}{1 + \tilde{r}}$. Importantly, the tax on initial assets acts as a lump-sum expense, $\tau_a A^i$, and

instead affects the savings and portfolio choices of investors by changing the effective returns on assets in the second period and the market interest rate, $\{\tilde{R}_2^i, \tilde{r}\}$.

Equation (4) lets us see clearly all the margins affected by the investors' choice of realizing gains: Selling a larger fraction of assets, ϕ^i , means more resources available for consumption in the first period at the cost of future returns $((1+\tilde{R}_2^i)/(1+\tilde{r})$ in the NPV of assets); moreover, taxation implies a tradeoff between paying τ_k on realized first period returns and paying $\tau_R + \frac{\tau_k}{1+\tilde{r}}$ if delaying realization. The first margin favors the realization of gains in period 1 as long as $\tilde{R}_2^i < \tilde{r}$. The second margin favors realization as long as $\tau_k < \left(\frac{1+\tilde{r}}{\tilde{r}}\right) \tau_R$. Without unrealized capital gains taxes, the discounting of deferred tax payments by $\tilde{r}$ generates an incentive for postponing capital gains.

2.3. Investors' problem

The key aspect of the investors' problem is that they face a classical *lock-in* effect: an investor may forgo a profitable investment if realizing capital gains is required to finance it. This deferral of capital gains need not be costly for investors, as they can finance consumption by borrowing funds rather than realizing capital gains, which coincides with the tax strategy of some of the wealthiest Americans (see, for instance, the leaked tax returns discussed in [Eisinger, Ernsthausen, and Kiel 2021](#)).

The problem of the investors is

$$V(A, R_1, R_2) = \max_{\{C_1, C_2, \phi\}} U(C_1, C_2) \quad \text{s.t. (4)} \quad (5)$$

The optimal choices are functions of the investor's characteristics, $X^*(A, R_1, R_2)$ for $X \in \{C_1, C_2, \phi\}$, that take as given common taxes and market returns. To avoid cumbersome notation, we use the shorthand X^i when referring to $X^*(A^i, R_1^i, R_2^i)$ from here onward.

The investors' optimal consumption and saving choice satisfies the Euler equation

$$\frac{U_{C_1}(C_1^i, C_2^i)}{U_{C_2}(C_1^i, C_2^i)} = 1 + \tilde{r}. \quad (6)$$

Despite the differences in private returns between investors, all investors have access to a common asset B , which serves as the marginal source of liquid funds. Hence, at the margin, all investors face the same relative price of period 2 vs period 1 consumption as reflected in the right-hand side of (6).

Investors also solve a portfolio problem when choosing how much of their assets to sell to realize their capital gains in period 1. Equation (7) describes the optimal choice:

$$\lambda^i \left(\frac{\tilde{r} - \tilde{R}_2^i}{1 + \tilde{r}} (1 + R_1^i) - \left(\frac{\tilde{r}\tau_k}{1 + \tilde{r}} - \tau_R \right) R_1^i \right) \leq 0, \quad (7)$$

where λ^i is the Lagrange multiplier of the intertemporal budget constraint (4). Here, the sign $\leq$ reflects whether it is optimal to retain or realize capital gains. When the left-hand side of (7) is positive, resources strictly increase in ϕ^i , and hence it is optimal for investors to fully liquidate their initial asset, $\phi^i = 1$. Conversely, when the left-hand side is negative, resources decrease in ϕ^i , and it is optimal for investors to keep their initial asset, $\phi^i = 0$ and use the risk-free asset B to finance consumption and tax payments.¹⁰

3. Portfolio choice and capital lock-in

We now derive our main result: wealth taxation enhances portfolio-choice efficiency while capital income taxes result in inefficient portfolio choice. The optimal portfolio choice described in (7) implies a “*reservation-return*” strategy for investors—a level for

¹⁰We ignore the knife-edge case in which the left-hand side equals zero since it holds for a measure zero of investors given our assumptions over the distribution of investors. See also Section 4.

R_2^i above which it is optimal to delay capital gains—with the reservation return given by

$$\bar{R}(R_1^i) \equiv r - \tau_k r \frac{R_1^i}{1 + R_1^i} + \frac{\tau_k \tau_a + (1 + (1 - \tau_k) r - \tau_a) \tau_R}{1 - \tau_k} \frac{R_1^i}{1 + R_1^i} \quad (8)$$

We emphasize the dependence of the second-period reservation return on the first-period return as taxes and the risk-free rate are common for all investors. Investors whose rate of return in the second period exceeds the reservation return, $R_2^i > \bar{R}(R_1^i)$, delay capital gains, setting $\phi^i = 0$. Investors whose rate of return lies below $\bar{R}(R_1^i)$ realize their gains by selling their assets, $\phi^i = 1$.

In the absence of taxes, the condition for delaying capital gains is $R_2^i > r$. This corresponds to the efficient allocation of resources, as only investors with returns below the market rate r realize their capital gains to invest in B and consume. It is also a special case of the more general principle of efficient portfolio choice in [Samuelson \(1969\)](#) and [Merton \(1969, 1971\)](#), where only the value of wealth and future returns are relevant states for the investor's choices. By contrast, realization-based capital income taxation makes past returns an additional state variable as they determine the tax-advantage of delaying realization ([Constantinides 1983](#); [Dammon, Spatt, and Zhang 2001](#)).

We therefore measure inefficiency in investors' portfolio choice by the presence of lock-in and lock-out. Specifically, an investor is *locked-in* if they delay capital gains, $\phi^i < 1$, despite having a return lower than the market rate, $R_2^i < r$; alternatively, an investor is *locked-out* if they realize their gains, $\phi^i > 0$, despite having a return higher than the market rate, $R_2^i > r$. In this way, locked-in and locked-out investors reduce total surplus from the loss in returns from their respective gaps between R_2 and r .

The characterization of the investors' optimal portfolio choice in (8) lets us describe the locked-in and locked-out investors in terms of the reservation return $\bar{R}$ and the

investors' returns (regardless of their wealth).¹¹ Formally,

Definition 1 (Locked-in and locked-out Investors).

$$\text{Lock-in} \iff \bar{R}(R_1) < R_2 < r ; \quad (9)$$

$$\text{Lock-out} \iff r < R_2 < \bar{R}(R_1) . \quad (10)$$

3.1. Taxation and inefficiency in portfolio choice

Here we see how the tax instruments interact with the investors' portfolio choice. Capital income taxes, τ_k , reduce $\bar{R}$, generating *capital lock-in* (second term in 8). That is, some investors delay capital gains even though their rate of return lies below the market return in order to avoid taxes upon realization of gains in the first period. By contrast, unrealized capital gains taxes, τ_R , increase $\bar{R}$ generating *capital lock-out*, with investors realizing capital gains and forgoing future returns. *Wealth taxes do not generate inefficient lock-in or lock-out on their own*, but do interact with the effects of other taxes.

We state these results formally in the following proposition and provide proofs for all results in Appendix B. For simplicity of exposition we discuss all the results assuming that returns are positive, but they extend naturally to negative returns by reversing the lock-in and lock-out effects as we illustrate in Appendix A.

Proposition 1 (Taxation and portfolio choice). *Assume $\tau_k, \tau_a, \tau_R \geq 0$.*

- (a) *If the capital income tax is zero, $\tau_k = 0$ and $\tau_a < 1 + r$, there is no capital lock-in.*
- (b) *A positive capital income tax, $\tau_k > 0$, in combination with zero wealth and unrealized capital gains taxes, $\tau_a = \tau_R = 0$, implies capital lock-in and no capital lock-out.*

¹¹In a multi-period setting, lock-in and lock-out correspond to cases in which the embedded or past capital gains determine realization, as opposed to realization depending only on the prospective future gains from holding onto the asset. We show in Appendix E that capital income taxes generate lock-in in this sense when investors face a multi-period portfolio problem with stochastic returns.

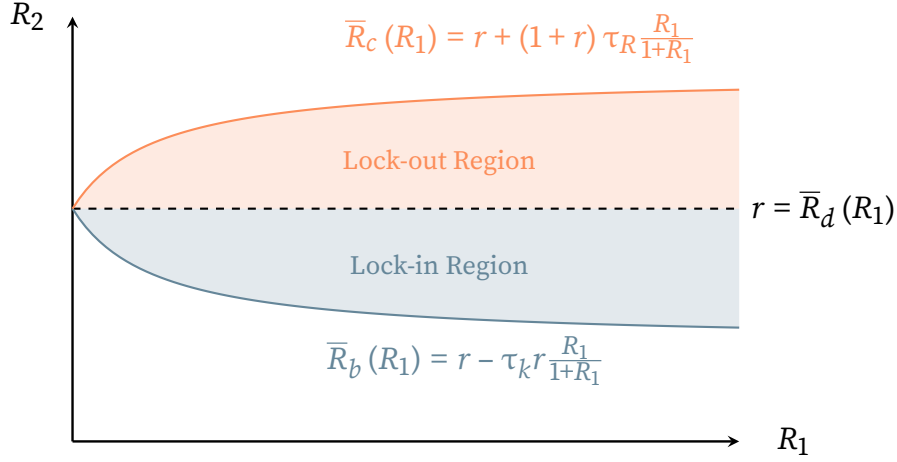
- (c) *A positive unrealized capital gains tax, $\tau_R > 0$, in combination with zero capital income and wealth taxes, $\tau_k = \tau_a = 0$, implies capital lock-out and no capital lock-in. Moreover, taxing all accrued gains by setting $\tau_R = \tau_k > 0$ also implies capital lock-out.*
- (d) *A positive wealth tax, $\tau_a > 0$, in combination with zero capital income and unrealized capital gains taxes, $\tau_k = \tau_R = 0$, generates no capital lock-in or lock-out.*

Proposition 1 shows that taxes on capital gains result in inefficient portfolio choices, either by locking capital in or out, as we show graphically in Figure 1. In the case of capital income taxes (on realized capital gains), τ_k , a higher tax rate incentivizes postponing gains to reduce the net-present value of tax payments. This generates capital lock-in, making $\bar{R}(R_1) < r$ as in part (b) of the proposition shown in blue in the figure. The reservation return decreases as returns grow and there is a larger incentive to delay tax payments, converging to $(1 - \tau_k)r$ as R_1^i approaches infinity. The area between the reservation return and the market return, r , represents the region where portfolio choice is distorted by lock-in. Increasing τ_k results in a larger fraction of inefficiently locked-in investors.

The opposite happens with taxes on unrealized capital gains, τ_R . These are also distortive but operate by locking out investors who sell “too early,” making $\bar{R}(R_1) > r$, as in part (c) of the proposition shown in red in the figure. As before, the shaded red area gives the region where portfolio choice is distorted by lock-out. The lock-out happens even when setting $\tau_R = \tau_k$ so as to tax all accrued returns, regardless of realization. So, taxing *accrued capital gains* distorts portfolio choice and is not a solution to capital lock-in and lock-out. The reason is that, while all returns are taxed in period one, unrealized returns are taxed again in period two, incentivizing lock-out.

Finally, while wealth taxes interact with the incentives for capital lock-in or lock-out, they do not generate them on their own. In fact, without taxes on capital gains (realized or otherwise) the wealth tax does not distort the portfolio allocation of investors as

FIGURE 1. Effect of Realized and Unrealized Capital Gains Taxation on Portfolio Choice



Notes: The figure illustrates the reservation return $\bar{R}(R_1)$ under the tax configurations in cases (b), (c), and (d) of Proposition 1. The blue downward-sloping curve corresponds to case (b) with a positive capital income tax, $\tau_k > 0$, and no wealth or unrealized capital gains taxes, $\tau_a = \tau_R = 0$. The shaded blue area contains investors locked into their asset. The red curve corresponds to case (c) with a positive unrealized capital gains tax, $\tau_R > 0$, and no capital income or wealth taxes, $\tau_k = \tau_a = 0$. The shaded red area contains investors locked out of their asset. The dashed line corresponds to case (d) with no capital income or unrealized capital gains taxes, $\tau_k = \tau_R = 0$, and no restrictions on wealth taxes. In this case, the reservation return is equal to the market return r .

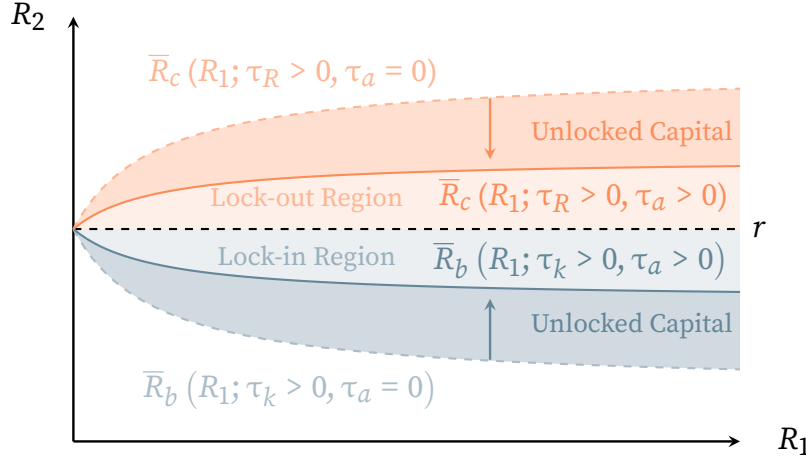
established in part (d) of the proposition and shown graphically by the dashed line in the figure. We turn to the interactions between wealth and capital income taxes next.

3.2. Unlocking capital

Capital income taxes distort investors' portfolio choices and can move the reservation return away from the efficient threshold r generating either capital lock-in or lock-out. However, wealth taxes can be used to restore efficiency to the investors' portfolio choice. Mechanically, τ_a pushes the reservation return $\bar{R}$ towards the market return r : upward when capital income taxes create lock-in, and downward when taxes on unrealized gains create lock-out, as shown in Figure 2. This effect can, under the conditions below, fully unlock capital.

Proposition 2 (Wealth taxes unlock capital). *Given capital gains taxes $0 \leq \tau_k < 1$ and*

FIGURE 2. Effect of Wealth Taxation on Portfolio Choice



Notes: The figure illustrates how a positive wealth tax changes the reservation return $\bar{R}(R_1)$ in the two tax configurations shown in Figure 1. In the lower half, $\tau_k > 0$ and $\tau_R = 0$. The dashed blue curve is $\bar{R}_b(R_1; \tau_k > 0, \tau_a = 0) = r - \tau_k r R_1 / (1 + R_1)$ and the solid blue curve is $\bar{R}_b(R_1; \tau_k > 0, \tau_a > 0) = r - \tau_k (r - \tau_a / (1 - \tau_k)) R_1 / (1 + R_1)$. In the upper half, $\tau_k = 0$ and $\tau_R > 0$. The dashed red curve is $\bar{R}_c(R_1; \tau_R > 0, \tau_a = 0) = r + (1 + r) \tau_R R_1 / (1 + R_1)$ and the solid red curve is $\bar{R}_c(R_1; \tau_R > 0, \tau_a > 0) = r + (1 + r - \tau_a) \tau_R R_1 / (1 + R_1)$. The darker areas between the curves are capital unlocked by the wealth tax.

$\tau_R \geq 0$, portfolio choice is efficient for all investors if and only if there are no capital gains taxes, $\tau_k = \tau_R = 0$, with $\tau_a \in [0, 1 + r)$, or capital gains taxes satisfy

$$0 \leq \tau_R \leq \frac{\tau_k (1 - \tau_k) r}{1 + (1 - \tau_k) r}; \quad \tau_k > 0, \quad (11)$$

and the wealth tax rate is

$$\tau_a = \tau_a^*(\tau_k, \tau_R) \equiv (1 - \tau_k) r - \frac{\tau_R}{\tau_k - \tau_R}. \quad (12)$$

If $\tau_k = \tau_R > 0$ there is capital lock-out for any wealth tax.

Proposition 2 provides a crisp characterization of how wealth taxes can unlock capital. Notably, the tax rate that achieves this does not depend on the (distribution of) locked-in capital gains $\{R_1^i\}$ and instead it only depends on the market rate r and other

tax instruments.¹² Further, this implies the wealth tax operates without needing to know the identity of investors. We show these results extend naturally to a multi-period setting with stochastic returns in Appendix E: the wealth tax in (12) removes embedded capital gains from the realization decision, leaving only future expected returns as the relevant margin. Moreover, it does so without changing the investor's risk exposure.

To see why setting the wealth tax rate at $\tau_a^*(\tau_k, \tau_R)$ eliminates both lock-in and lock-out, substitute (12) into the final term of the intertemporal budget constraint (4), which gives the net present value of all taxes paid on the first-period return R_1^i :

$$\left(\left(\tau_R + \frac{\tau_k}{1 + (1 - \tau_k) r - \tau_a^*(\tau_k, \tau_R)} \right) (1 - \phi^i) + \tau_k \phi^i \right) R_1^i A^i = \tau_k R_1^i A^i. \quad (13)$$

Under $\tau_a^*(\tau_k, \tau_R)$, the net present value of tax payments does not depend on the portfolio choice ϕ^i , and instead sets the tax bill equal to the capital income tax on accrued gains (regardless of realization!), resulting in efficient portfolio choice. Intuitively, both lock-in and lock-out are the result of investors attempting to reduce their period-1 tax payments: in the case of lock-in, by holding on to the asset to defer the capital-gains tax; in the case of lock-out, by selling it to avoid the tax on unrealized gains. The wealth tax counteracts both forces because its period-2 base falls one-for-one with any tax paid in period 1, so delaying tax payments to reduce capital-gains liabilities raises the wealth tax owed later. At $\tau_a^*(\tau_k, \tau_R)$ these two effects exactly offset, delivering the tax-neutrality result in (13).

A complementary way to understand the relationship between the tax instruments emerges when considering three corner cases that cover the incentives for lock-in and lock-out induced by capital gains taxation. The wealth tax interacts with the incentives from other taxes by altering the effective discount rate on future tax payments through its effect on the after-tax market rate $\tilde{r}$.

¹²Intuitively, the tax benefits from delaying capital gains are proportional to R_1^i , as is the increase in wealth tax payments in period 2 implied by the delay. So, the same wealth-tax rate balances these terms for all investors, independent of their returns.

Corollary 1. *Proposition 2 implies the following corner cases:*

- (a) **No *unrealized capital gains tax*:** *If $\tau_k > 0$ and $\tau_a = \tau_R = 0$ there is capital lock-in. The wealth tax that eliminates lock-in is $\tau_a^*(\tau_k, 0) = (1 - \tau_k) r$*
- (b) **No *capital income tax*:** *If $\tau_R > 0$ and $\tau_a = \tau_k = 0$ there is capital lock-out. A wealth tax can only eliminate lock-out by confiscating all market returns, that is $\tau_a^*(0, \tau_R) = 1 + r$.*
- (c) **No *wealth tax*:** *If $\tau_a = 0$, efficiency in investors' portfolio choice requires a tax on capital gains that exactly cancels out the lock-in and lock-out incentives by setting*

$$\tau_R = \frac{\tau_k (1 - \tau_k) r}{1 + (1 - \tau_k) r}. \quad (14)$$

The first case in Corollary 1 shows how wealth taxes undo capital lock-in. A capital income tax creates a tax-deferral benefit as future tax payments are discounted. This makes the present value of the tax payments lower according to the discounting of the effective market rate $\tilde{r}$, as seen in the intertemporal budget constraint (4). The result is a negative wedge in the optimal portfolio choice, seen in the second term of the reservation return (8). Wealth taxes work by lowering the after-tax market return $\tilde{r}$ (see also Guvenen et al. 2023, 2024). At $\tau_a = (1 - \tau_k) r$ we have $\tilde{r} = 0$, so future tax payments are no longer discounted. This eliminates the tax advantage of delaying capital gains.¹³

While wealth taxes can undo capital lock-in, they can only undo the capital lock-out generated by taxing unrealized capital gains if wealth were to be taxed at an infeasibly high rate, as stated in the second case in the Corollary. A tax on unrealized capital gains directly penalizes holding the asset, with some investors opting for the market return instead. This incentive can only be negated if the market return were to be confiscated along with the principal of the investment. This establishes a novel margin in which

¹³Setting the wealth tax above $\tau_a^*(\tau_k, \tau_R)$ results in inefficient lock-out and setting it below maintains lock-in, but only in the presence of capital income taxes (Proposition 1).

taxing unrealized capital gains is more distortive than taxing realized capital gains.

The third case shows how the lock-in and lock-out incentives of capital gains taxes balance. It also provides an explanation for the restriction in (11) imposed in Proposition 2. A wealth tax can restore efficiency only when the unrealized capital gains tax is not too large relative to the capital income tax. The upper bound in (11) is exactly the unrealized capital gains tax that would offset the lock-in incentive in the absence of a wealth tax. At that upper bound, the required wealth tax is zero. Above the upper bound but below τ_k , the formula in (12) would require a negative wealth tax, reflecting that the lock-out incentives from τ_R are stronger than the lock-in incentives from τ_k .

4. Optimal taxes

Up to now, we have focused on the efficiency of the investors' portfolio choice, leaving aside the intertemporal distortion from capital gains and wealth taxes. However, setting positive capital income taxes and wealth taxes is bound to distort intertemporal choices for investors regardless of their discount factors and returns. Therefore, when capital income taxes are positive, the optimal tax mix must trade off the intertemporal distortion against the portfolio-efficiency gains from reducing lock-in or lock-out. We now study this trade-off when the government chooses welfare-maximizing tax rates. We set $\tau_R = 0$ in order to focus on the main trade-off between intertemporal and portfolio distortions already present in τ_k and τ_a . We provide results with τ_R in Appendix C.

4.1. The government's problem

The government chooses taxes to maximize a utilitarian social welfare function:

$$\mathbb{V} \equiv \int \alpha^i V(A^i, R_1^i, R_2^i) di, \quad (15)$$

where α^i denotes the Pareto weight on investor i , which captures the government's redistributive preferences. The distribution of (A^i, R_1^i, R_2^i) is described by the cumulative distribution function $F(A^i, R_1^i, R_2^i)$ with domain given by $A^i \in \Omega \subset \mathbb{R}_+$ and $(R_1^i, R_2^i) \in [R^{\min}, R^{\max}]^2$. We assume F is absolutely continuous. Finally, we also assume that the government does not discriminate between individuals with the same assets and returns, so that the Pareto weights are $\alpha^i = \alpha(A^i, R_1^i, R_2^i)$. Accordingly, we drop the index i below whenever possible.

The government collects taxes to finance lump-sum transfers T and exogenous expenditure whose net present value equals G . The government can lend or borrow at the market interest rate r and faces a budget constraint

$$G + T \leq \int [\tau_k \mathcal{J}(A, R_1, R_2) + \tau_a \mathcal{A}(A, R_1, R_2)] dF(A, R_1, R_2), \quad (16)$$

where $\mathcal{J}(\cdot)$ is the net present value of taxable capital income,

$$\mathcal{J}(A, R_1, R_2) \equiv \phi^*(A, R_1, R_2) R_1 A + \frac{(1 - \phi^*(A, R_1, R_2))(R_1 + R_2(1 + R_1))A + rB^*(A, R_1, R_2)}{1 + r} \quad (17)$$

and $\mathcal{A}(\cdot)$ is the net present value of the wealth tax base,

$$\mathcal{A}(A, R_1, R_2) \equiv A + \frac{(1 - \phi^*(A, R_1, R_2))(1 + R_1)A + B^*(A, R_1, R_2)}{1 + r}. \quad (18)$$

All quantities are evaluated at the investors' optimal choices under the tax system and discounted using the market's rate r . The government chooses τ_k , τ_a , and T subject to lower-bound constraints $\tau_k, \tau_a \geq 0$ and $T \geq T_0$, together with the admissibility requirements $\tau_k < 1$, $\tau_a < 1$, and $1 + \tilde{r} = 1 + (1 - \tau_k)r - \tau_a > 0$.

4.2. The response of tax bases to tax rates

In order to solve for optimal taxes we must first account for the discontinuity of portfolio choices at the reservation return $R_2^i = \bar{R}(R_1^i)$. At this cutoff, taxable income $\mathcal{J}(\cdot)$ and wealth $\mathcal{A}(\cdot)$ jump, depending on whether investors sell or keep their initial asset. These jumps determine discrete changes in tax bases when the reservation return changes (for instance as tax rates change). The following lemma defines these jumps.

Lemma 1. Assume $\tau_R = 0$, $\tau_k, \tau_a \geq 0$, $\tau_k < 1$, and $1 + \tilde{r} > 0$.

The changes in tax bases $\mathcal{J}(\cdot)$ and $\mathcal{A}(\cdot)$ at the reservation return are

$$\Delta \mathcal{J}(A, R_1) \equiv \lim_{R_2 \xrightarrow{\bar{R}} \bar{R}(R_1)} \mathcal{J}(A, R_1, R_2) - \lim_{R_2 \xrightarrow{+} \bar{R}(R_1)} \mathcal{J}(A, R_1, R_2) = \left(r - \frac{\tau_k \tau_a}{1 - \tau_k} \right) \frac{R_1 A}{1 + r}, \quad (19)$$

$$\Delta \mathcal{A}(A, R_1) \equiv \lim_{R_2 \xrightarrow{\bar{R}} \bar{R}(R_1)} \mathcal{A}(A, R_1, R_2) - \lim_{R_2 \xrightarrow{+} \bar{R}(R_1)} \mathcal{A}(A, R_1, R_2) = -\tau_k \frac{R_1 A}{1 + r}. \quad (20)$$

The marginal seller reports more taxable capital income than the marginal holder because selling accelerates realization ($\Delta \mathcal{J} > 0$). Conversely, the resulting capital-income-tax payment mechanically reduces the wealth carried into period 2, so the seller also reports less taxable wealth ($\Delta \mathcal{A} < 0$ when $\tau_k > 0$).

Applying the respective rates to (19) and (20) gives the total tax revenue jump:

$$\tau_k \Delta \mathcal{J} + \tau_a \Delta \mathcal{A} = \tau_k \left(r - \frac{\tau_a}{1 - \tau_k} \right) \frac{R_1 A}{1 + r}. \quad (21)$$

The right-hand side of (21) shows how the two taxes interact: realization raises capital-income-tax revenue but simultaneously erodes the wealth-tax base, since the income-tax payment reduces the wealth carried forward (as in the discussion of Proposition 2). This cross-base effect is captured by $\tau_a/(1 - \tau_k)$ —the fiscal gain from realization is smaller than $\tau_k r$ because part of the income-tax revenue is offset by a loss in wealth-tax receipts.

At $\tau_a^* = (1 - \tau_k)r$, the total revenue jump (21) equals zero: the additional income-tax

revenue from realization is exactly offset by the reduction in the wealth-tax base. The government is therefore fiscally indifferent between the marginal investor selling or holding—precisely the condition for portfolio efficiency. This complements the intuition from Corollary 1: setting $\tilde{r} = 0$ eliminates the discounting advantage of deferral, and the revenue-neutrality condition here is simply the other side of that same coin.

Using Lemma 1 we can express the optimality conditions of the government's problem in terms of the elasticities of the tax bases $\mathcal{I}$ and $\mathcal{A}$ to τ_k and τ_a and the welfare weights (as in Saez and Stantcheva 2016). Welfare weights, g^i , are defined as

$$g^i \equiv g(A^i, R_1^i, R_2^i) \equiv \frac{\alpha^i U_{C_1}}{\eta}, \quad (22)$$

where η is the multiplier on the government budget constraint (16). This definition measures the direct welfare value of transferring resources to investor i .

The relevant elasticities shaping the optimal tax choice capture how the tax base responds to investors' savings and portfolio choices. First, when the tax rate changes (either for capital income or wealth taxes), there is a change in tax base coming from the intertemporal choice of the risk-free asset B^i . This is captured by the (uncompensated) intertemporal elasticity of tax base $\mathcal{Y}$ with respect to the net-of-tax rate $1 - \tau$ as

$$\varepsilon_{\mathcal{Y}, 1-\tau}(A, R_1, R_2) \equiv -\frac{\partial \mathcal{Y}}{\partial \tau} \frac{1-\tau}{\mathcal{Y}}, \quad \mathcal{Y} \in \{\mathcal{I}, \mathcal{A}\}, \quad \tau \in \{\tau_k, \tau_a\}. \quad (23)$$

Second, there is also a change in the tax base coming from the portfolio choice, as the reservation return $\bar{R}$ changes and marginal investors switch between selling and holding onto their assets. This is captured by the portfolio elasticity

$$\xi_{\mathcal{Y}, 1-\tau}(A, R_1) \equiv -\frac{\frac{\partial \bar{R}(R_1)}{\partial \tau} f^{R_2}(\bar{R}(R_1)|A, R_1) \Delta \mathcal{Y}(A, R_1) (1-\tau)}{\mathbb{E}[\mathcal{Y}(\cdot) | A, R_1]}, \quad \mathcal{Y} \in \{\mathcal{I}, \mathcal{A}\}, \quad \tau \in \{\tau_k, \tau_a\}. \quad (24)$$

The portfolio elasticity takes into account the jump in the tax base, $\Delta\mathcal{Y}$, which occurs for marginal investors (those with $R_2^i = \bar{R}(R_1)$). Accordingly, the elasticity takes into account the density of marginal investors, $f^{R_2}(\bar{R}(R_1)|A, R_1)$, where f^{R_2} is the conditional density of R_2 given (A, R_1) . The expectation in the denominator is over the conditional distribution F^{R_2} .

4.3. Sufficient statistics for optimal taxes

We can now state the conditions for optimal taxation in terms of sufficient statistics:

Proposition 3 (Optimal taxes). *Optimal taxes $\{\tau_k, \tau_a, T\}$ satisfy*

$$1 - \bar{g}^J \leq \frac{\tau_k}{1 - \tau_k} \left(\bar{\varepsilon}_{J,1-\tau_k}^J + \bar{\xi}_{J,1-\tau_k}^J \right) + \frac{\tau_a}{1 - \tau_k} \frac{\bar{A}}{\bar{J}} \left(\bar{\varepsilon}_{A,1-\tau_k}^A + \bar{\xi}_{A,1-\tau_k}^A \right), \quad (25)$$

$$1 - \bar{g}^A \leq \frac{\tau_a}{1 - \tau_a} \left(\bar{\varepsilon}_{A,1-\tau_a}^A + \bar{\xi}_{A,1-\tau_a}^A \right) + \frac{\tau_k}{1 - \tau_a} \frac{\bar{J}}{\bar{A}} \left(\bar{\varepsilon}_{J,1-\tau_a}^J + \bar{\xi}_{J,1-\tau_a}^J \right), \quad (26)$$

$$1 - \bar{g} \geq \mathbb{E} \left[\tau_k \frac{\partial J}{\partial T} + \tau_a \frac{\partial A}{\partial T} \right]. \quad (27)$$

Here, $\mathbb{E}[x(\cdot)]$ denotes the population average of $x(\cdot)$, $\bar{x}^{\mathcal{Y}} \equiv \mathbb{E}[x(\cdot) \mathcal{Y}(\cdot)] / \mathbb{E}[\mathcal{Y}(\cdot)]$ denotes the average weighted by tax base $\mathcal{Y} \in \{J, A\}$, and $\frac{\partial \mathcal{Y}}{\partial T}$ denotes the change in tax base $\mathcal{Y}$ that results from a change in the lump-sum transfer (i.e. the income effect).

Proposition 3 shows the tradeoffs faced by the government when setting taxes. The left-hand sides of (25)–(27) are the mechanical welfare gains from raising τ_k , τ_a , and T , respectively. The right-hand sides of (25) and (26) are the marginal excess burdens per unit of revenue. Crucially, τ_k and τ_a generate an *own-base* and a *cross-base* response. These responses operate through changes in savings and portfolio choices captured by the elasticities defined in (23) and (24). The cross-base response is scaled by the relative size of the cross base, reflecting that this effect becomes more important when the cross-base is larger relative to the own-base.

The cross-base response is key because it captures how the lock-in incentives of both taxes push in opposite directions. Specifically, for relevant values of taxes, the cross-base portfolio elasticities are negative ($\bar{\xi}_{J,1-\tau_a}, \bar{\xi}_{A,1-\tau_k} < 0$). Intuitively, increasing τ_k reduces the share of investors who sell their assets, decreasing the reservation return, reducing taxable income but increasing taxable wealth. Conversely, increasing τ_a increases the share of investors who sell their assets, increasing taxable income but reducing taxable wealth. By contrast, the portfolio own-base elasticities are positive ($\bar{\xi}_{J,1-\tau_k}, \bar{\xi}_{A,1-\tau_a} > 0$) as are the own-base responses to taxes ($\bar{\varepsilon}_{y,1-\tau} > 0$)—as a change in τ_k or τ_a reduces the effective market rate $\tilde{r}$, discouraging savings. These effects crystallize the efficiency trade-offs from wealth taxation: The wealth tax is costly on the intertemporal margin (own-base) but can be beneficial on the portfolio margin when realization-based capital income taxes create lock-in (cross-base).

Subtracting (26) from (25) gives a condition that balances the *marginal excess burden* of capital income and wealth taxes with the relative welfare weights on investors holding capital income and wealth:

$$\text{MEB}_k = \text{MEB}_a + (\bar{g}^A - \bar{g}^J) . \quad (28)$$

The marginal excess burdens collect the effects of each tax on government revenue through behavioral changes in savings and portfolio choices. They correspond to the right-hand-sides of (25) and (26). A positive marginal burden reflects tax-base erosion, while a negative burden reflects tax-base expansion or a corrective portfolio effect. All else equal, if $\text{MEB}_k > \text{MEB}_a$, a revenue-neutral reform that reduces τ_k and raises τ_a increases economic efficiency, and vice versa. We provide further discussion of the components of the marginal excess burdens in Appendix B.1.

The right-hand side of (27) captures the fiscal externality of the lump-sum transfer. A higher transfer raises period-1 resources, expanding the income and wealth tax bases

($\frac{\partial \mathcal{U}}{\partial T} > 0$ for $\mathcal{U} \in \{\mathcal{J}, \mathcal{A}\}$). This positive fiscal externality is of the opposite sign to those of τ_k and τ_a , explaining why the inequality in (27) is reversed relative to (25)–(26).¹⁴

4.4. A sufficient condition for wealth taxation

One important implication of Proposition 3 is that it gives a sufficient condition for the desirability of wealth taxes in the presence of capital income taxes. Crucially, the condition relies on efficiency considerations for a positive wealth tax instead of redistributive motives.¹⁵ To see this, fix $\tau_k > 0$ and consider condition (26) in Proposition 3. The optimal wealth tax is positive if, evaluated at $\tau_a = 0$, the benefits of the first unit of wealth taxation exceed its marginal excess burden, that is,

$$1 - \bar{g}^{\mathcal{A}} > \tau_k \bar{\mathcal{A}}^{\bar{\mathcal{J}}} \left(\bar{\varepsilon}_{\mathcal{J}, 1-\tau_a}^{\mathcal{J}} + \bar{\xi}_{\mathcal{J}, 1-\tau_a}^{\mathcal{J}} \right). \quad (29)$$

A positive wealth tax is therefore more likely when the redistributive value of the wealth tax is high ($\bar{g}^{\mathcal{A}}$ is small), or when the cross-elasticity of taxable capital income with respect to the wealth tax is negative, combining a small intertemporal cross-elasticity ($\bar{\varepsilon}_{\mathcal{J}, 1-\tau_a}^{\mathcal{J}}$) and a strongly negative portfolio cross-elasticity ($\bar{\xi}_{\mathcal{J}, 1-\tau_a}^{\mathcal{J}}$). This last condition turns out to be sufficient for the wealth tax to be positive.

Corollary 2 (Sufficient condition for wealth taxes). *Set $\tau_k > 0$. If the government weakly values redistribution away from the wealth tax base such that pure welfare weights satisfy*

¹⁴One can rewrite (27) as $\mathbb{E} \left[g^i \right] \leq 1 - \mathbb{E} \left[\tau_k \frac{\partial \mathcal{J}}{\partial T} + \tau_a \frac{\partial \mathcal{A}}{\partial T} \right] \leq 1$, confirming that the left-hand sides of all three first-order conditions are non-negative whenever the government is weakly inequality averse — that is, whenever social welfare weights $g(\cdot)$ are non-increasing in taxable wealth, so that $1 \geq \bar{g} \geq \bar{g}^{\mathcal{J}}, \bar{g}^{\mathcal{A}}$.

¹⁵Most previous research that calls for a positive wealth tax (as in Piketty 2013; Saez and Zucman 2019) as well as optimal wealth-tax models (like those in Scheuer and Slemrod 2021; Piketty, Saez, and Zucman 2023), justify a positive wealth tax from the perspective of reducing inequality in the (initial) wealth distribution. Notable exceptions are Guvenen et al. (2023) and Guvenen et al. (2024), who look at the efficiency gains from capital reallocation and innovation produced by book-value wealth taxes.

$\bar{g}^A \leq \mathbb{E}[g(\cdot)]$, then a sufficient condition for a positive wealth tax is

$$\bar{\varepsilon}_{j,1-\tau_a}^j + \bar{\xi}_{j,1-\tau_a}^j < 0, \quad (30)$$

Corollary 2 shows a positive wealth tax may be desirable even if the government is indifferent with respect to wealth inequality, $\bar{g}^A = \mathbb{E}[g(\cdot)]$. Further, Equation (30) provides a clear empirical test for the desirability of a wealth tax. The relevant statistic is the cross-elasticity of taxable capital income with respect to the wealth tax, not the own-elasticity of taxable wealth that much of the empirical literature has focused on (see, among others, Seim 2017; Jakobsen et al. 2020; Londoño-Vélez and Ávila-Mahecha 2021; Ring 2024). Intuitively, for small τ_a , own-base distortions vanish even if elasticities are large, whereas cross-base responses remain first-order.¹⁶

Effectively, what matters is the magnitude and responsiveness of capital lock-in and not the revenues coming directly from the wealth tax. If the wealth tax is effective at unlocking capital, the gains in the portfolio efficiency margin dominate the costs on investors' savings choices, even when the government is neutral about wealth redistribution and regardless of the revenues coming directly from wealth taxation.

To our knowledge, the closest empirical evidence is Lefèbvre and Zanoutene (2022), who estimate the effect of dividend taxation on the wealth-tax base in France and find a negative cross-elasticity. Interpreting dividend taxation as capital-income taxation, this finding is consistent with the mechanism we emphasize: higher capital-income taxes increase lock-in and expand the wealth-tax base, suggesting that a wealth tax that unlocks capital may conversely expand the capital-income tax base. Condition (30), however, calls for the reverse cross-elasticity — the response of the capital-income tax base to the wealth-tax rate — which to our knowledge has not yet been directly

¹⁶A similar condition applies for unrealized capital gains taxes. The trade-off between portfolio choice correction and intertemporal distortions is present in general, but only the portfolio component is present in our 2-period setup where τ_R does not directly affect $\tilde{r}$. See Appendix C for further discussion.

estimated. Our sufficient-statistic approach thus delivers a precise empirical target: estimating this single cross-elasticity would be enough to establish whether a wealth tax is desirable on efficiency grounds alone, independent of redistributive motives.

5. The design of wealth taxes and capital lock-in

We now discuss how the design of the wealth tax interacts with the lock-in incentives from capital income taxes. We have assumed so far that the wealth tax is levied on the historical value of wealth, namely the principal of the investment in each period. However, most proposals for wealth taxes and most implementations of the tax across countries levy it on the market value of wealth. This distinction on the tax base matters for how wealth taxes affect investors.

Specifically, a market-value wealth tax combines two conceptually different instruments: a tax on the historical value of wealth and a tax on accrued capital gains.¹⁷ To see this, consider a market-value wealth tax at rate τ_m in period 1 that raises

$$\underbrace{\tau_m W_1^i}_{\text{market-value wealth tax}} = \underbrace{\tau_m A^i}_{\text{historical wealth tax}} + \underbrace{\tau_m R_1^i A^i}_{\text{accrued capital gains tax}}. \quad (31)$$

The tax on accrued capital gains leads to inefficient double taxation. This tax is paid whether or not gains are realized. Then, gains are taxed again by τ_k when they are eventually realized. Ultimately, the two instruments work to lower investors' returns and imply an effective return $\tilde{r}_m \equiv (1 - \tau_k - \tau_m) r - \tau_m$. The double taxation is now apparent as returns are taxed by τ_k and τ_m .

We show in Appendix D that this double taxation lowers the wealth tax rate that restores portfolio efficiency, which is achieved under $\tau_m^* = \frac{(1-\tau_k)r}{1+r} < \tau_a^*$. Efficiency

¹⁷Alternatively, the wealth tax is calculated on end-of-period wealth, but payable in the next period, as is common in their implementation (see, for example, [Thoresen, Ring, Nygård, and Epland 2022](#)).

is restored under both tax designs by reducing the effective market return to zero, eliminating the tax advantage of delaying capital gains. However, a market-value wealth tax bundles a historical wealth tax with a tax on accrued capital gains, and thus operates on two margins simultaneously.¹⁸

An alternative design is to tax the value of the original principal, without updating its value after the first period. This design freezes the tax base of wealth taxes until new investments are made and has the advantage of not requiring any information on capital gains. Accordingly, taxing only the original principal distorts savings choices less (say, relative to the market value wealth tax we just discussed). Nevertheless, we show in Appendix D that divorcing wealth taxes from returns generates an additional source of capital lock-in. It incentivizes investors to delay capital gains to keep their wealth tax base low with the wealth tax increasing rather than undoing lock-in.

6. Conclusions

Realization-based capital income taxation can lock investors into assets whose returns are below the market return by making tax deferral valuable. A historical wealth tax can undo this distortion because it reduces the effective return used to discount future tax liabilities, and at the right rate it restores efficient portfolio choice without requiring investor-specific tax rates. The interplay of wealth taxes with capital lock-in introduces a new margin to the design of optimal taxes. This margin depends on cross-base responses: a wealth tax is desirable when unlocking capital expands the capital-income tax base enough to offset its savings distortion. Thus, the case for wealth taxation depends not only on equity weights, but also on how tax bases respond across instruments and the degree of capital lock-in.

¹⁸In economies with imperfect financial markets, investors differ in their marginal returns and the tax on accrued capital gains generates an additional distortion as it falls on high-return investors who are likely to be more productive and constrained, worsening misallocation. A historical-value wealth tax avoids this by not taxing current returns, similarly to the wealth tax in [Guvenen et al. \(2023, 2024\)](#).

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Appendix A. Lock-in and lock-out reversed

We implicitly assume that returns are positive in the main text, with $R^{\min} \geq 0$. This is done for simplicity of exposition. The reservation-return formula in (8) also characterizes portfolio choice following an initial loss, $-1 < R_1 < 0$, with the signs of the realization wedges reversed relative to those in Section 3. With $\tau_k > 0$ and no other taxes,

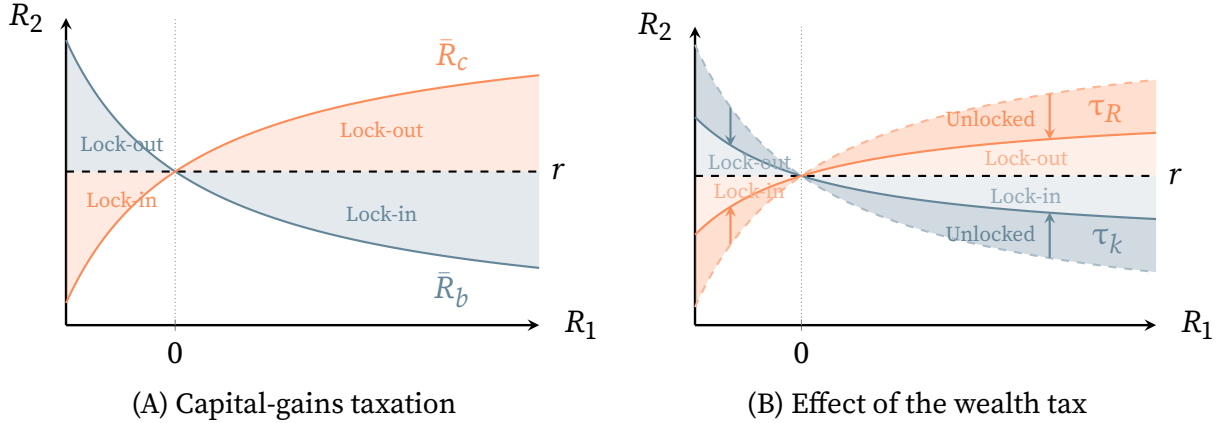
$$\bar{R}_b(R_1) - r = -\tau_k r \frac{R_1}{1 + R_1} > 0, \quad (\text{A1})$$

which produces lock-out rather than lock-in. Realizing the loss generates an immediate tax benefit, inducing some investors to sell even when their future private return exceeds the market return. Conversely, there is lock-in with only $\tau_R > 0$,

$$\bar{R}_c(R_1) - r = (1 + r)\tau_R \frac{R_1}{1 + R_1} < 0. \quad (\text{A2})$$

A historical wealth tax generates neither lock-in nor lock-out on its own as in Proposition 1, with $\bar{R}(R_1) = r$. The wealth tax changes the intertemporal return on liquid wealth, but it does not make realization depend on whether the private asset has accumulated a gain or a loss.

FIGURE A1. Taxation and Portfolio Choice with Positive and Negative Initial Returns



Notes: Panel A extends Figure 1 to initial returns $-1 < R_1 < 0$. The blue curve is $\bar{R}_b(R_1) = r - \tau_k r R_1 / (1 + R_1)$ and the red curve is $\bar{R}_c(R_1) = r + (1 + r)\tau_R R_1 / (1 + R_1)$. For negative initial returns, the blue region is lock-out and the red region is lock-in; these classifications reverse when $R_1 > 0$. Panel B extends Figure 2. Dashed curves set $\tau_a = 0$, while solid curves set $\tau_a > 0$ below the portfolio-efficient rate.

Appendix B. Proofs

This appendix contains the proofs for all results in the main text. We repeat the statement of the results for the reader's convenience.

Proposition A1 (Taxation and portfolio choice). *Assume $\tau_k, \tau_a, \tau_R \geq 0$.*

- (a) *If the capital income tax is zero, $\tau_k = 0$ and $\tau_a < 1 + r$, there is no capital lock-in.*
- (b) *A positive capital income tax, $\tau_k > 0$, in combination with zero wealth and unrealized capital gains taxes, $\tau_a = \tau_R = 0$, implies capital lock-in and no capital lock-out.*
- (c) *A positive unrealized capital gains tax, $\tau_R > 0$, in combination with zero capital income and wealth taxes, $\tau_k = \tau_a = 0$, implies capital lock-out and no capital lock-in. Moreover, taxing all accrued gains by setting $\tau_R = \tau_k > 0$ also implies capital lock-out.*
- (d) *A positive wealth tax, $\tau_a > 0$, in combination with zero capital income and unrealized capital gains taxes, $\tau_k = \tau_R = 0$, generates no capital lock-in or lock-out.*

PROOF. The results follow from replacing the specified tax rates into (8) and comparing $\bar{R}$ with r . When $R_1 = 0$ we have $\bar{R} = r$ always. We now consider the cases in which $R_1 > 0$.

- (a) Setting $\tau_k = 0$ yields $\bar{R}(R_1) = r + (1 + r - \tau_a) \tau_R \frac{R_1}{1 + R_1} \geq r$ for all $R_1 \geq R^{\min}$, precluding lock-in.
- (b) Setting $\tau_a = \tau_R = 0$ yields $\bar{R}(R_1) = r - \tau_k r \frac{R_1}{1 + R_1} < r$ for all $R_1 \geq R^{\min}$, ensuring lock-in.
- (c) Setting $\tau_k = \tau_a = 0$ yields $\bar{R}(R_1) = r + (1 + r) \tau_R \frac{R_1}{1 + R_1} > r$ for all $R_1 \geq R^{\min}$, ensuring lock-out. Setting $\tau_R = \tau_k > 0$ gives $\bar{R}(R_1) = r + \frac{\tau_R}{1 - \tau_R} \frac{R_1}{1 + R_1} > r$.
- (d) Setting $\tau_k = \tau_R = 0$ yields $\bar{R}(R_1) = r$ for all R_1 , ensuring efficiency.

□

Proposition A2 (Wealth taxes unlock capital). *Given capital gains taxes $0 \leq \tau_k < 1$ and $\tau_R \geq 0$, portfolio choice is efficient for all investors if and only if there are no capital gains taxes, $\tau_k = \tau_R = 0$, with $\tau_a \in [0, 1 + r)$, or capital gains taxes satisfy*

$$0 \leq \tau_R \leq \frac{\tau_k (1 - \tau_k) r}{1 + (1 - \tau_k) r}; \quad \tau_k > 0, \quad (\text{A3})$$

and the wealth tax rate is

$$\tau_a = \tau_a^*(\tau_k, \tau_R) \equiv (1 - \tau_k) r - \frac{\tau_R}{\tau_k - \tau_R}. \quad (\text{A4})$$

If $\tau_k = \tau_R > 0$ there is capital lock-out for any wealth tax.

PROOF. Portfolio choice is efficient for all investors if and only if the reservation return coincides with the market return for all $R_1 > 0$. Using (8), this condition is

$$0 = \bar{R}(R_1) - r = \left[-\tau_k r + \frac{\tau_k \tau_a + (1 + (1 - \tau_k) r - \tau_a) \tau_R}{1 - \tau_k} \right] \frac{R_1}{1 + R_1}. \quad (\text{A5})$$

In order to satisfy the condition the term in square brackets must equal zero. This is immediate if $\tau_k = \tau_R = 0$. When $\tau_k \neq \tau_R$ and at least one tax is positive we rearrange (A5) and solve for τ_a to obtain (12). For this value to be admissible it must also be that $1 + \tilde{r} > 0$. Therefore, when the wealth tax is set equal to $\tau_a^*(\tau_k, \tau_R)$,

$$1 + \tilde{r} = 1 + (1 - \tau_k) r - \tau_a^*(\tau_k, \tau_R) = \frac{\tau_k}{\tau_k - \tau_R}. \quad (\text{A6})$$

Thus, the requirement $1 + \tilde{r} > 0$ implies $\tau_k > 0$ and $\tau_R < \tau_k$. Under this condition, the denominator of (12) is positive, so the non-negativity requirement $\tau_a^*(\tau_k, \tau_R) \geq 0$ is equivalent to (11). This bound also implies $\tau_R < \tau_k$ because $(1 - \tau_k) r / (1 + (1 - \tau_k) r) < 1$. Hence (11) is necessary and sufficient for the wealth tax rate in (12) to be nonnegative and consistent with $1 + \tilde{r} > 0$. Finally, if $\tau_k = \tau_R > 0$, the term in square brackets in (A5) simplifies to $\tau_k / (1 - \tau_k) > 0$ and is independent of τ_a , so there is capital lock-out and no wealth tax can restore efficiency. If $\tau_k = \tau_R = 0$, the wedge is zero for all R_1 and all wealth tax rates that are nonnegative and preserve $1 + \tilde{r} > 0$. $\square$

Lemma A1. Assume $\tau_R = 0$, $\tau_k, \tau_a \geq 0$, $\tau_k < 1$, and $1 + \tilde{r} > 0$.

The changes in tax bases $\mathcal{J}(\cdot)$ and $\mathcal{A}(\cdot)$ at the reservation return are

$$\Delta \mathcal{J}(A, R_1) \equiv \lim_{R_2 \rightarrow \bar{R}(R_1)} \mathcal{J}(A, R_1, R_2) - \lim_{R_2 \xrightarrow{+} \bar{R}(R_1)} \mathcal{J}(A, R_1, R_2) = \left(r - \frac{\tau_k \tau_a}{1 - \tau_k} \right) \frac{R_1 A}{1 + r}, \quad (\text{A7})$$

$$\Delta \mathcal{A}(A, R_1) \equiv \lim_{R_2 \rightarrow \bar{R}(R_1)} \mathcal{A}(A, R_1, R_2) - \lim_{R_2 \xrightarrow{+} \bar{R}(R_1)} \mathcal{A}(A, R_1, R_2) = -\tau_k \frac{R_1 A}{1 + r}. \quad (\text{A8})$$

PROOF. Start with the two tax bases in (17) and (18). It is useful to write them in terms of the optimal first-period consumption $C_1(\cdot)$. If the marginal investor sells, $\phi = 1$, then the period-1 budget implies their B is

$$B^s = (1 + R_1)A + T - C_1 - \tau_k R_1 A - \tau_a A.$$

If the marginal investor holds the asset, $\phi = 0$, then their B is

$$B^h = T - C_1 - \tau_a A.$$

Substituting these expressions into the tax bases gives the expressions for sellers and keepers

$$\mathcal{J}^s = R_1 A + r \frac{(1+R_1)A + T - C_1 - \tau_k R_1 A - \tau_a A}{1+r}; \quad \mathcal{J}^h = \frac{[R_1 + R_2(1+R_1)] A + r(T - C_1 - \tau_a A)}{1+r}; \quad (\text{A9})$$

$$\mathcal{A}^s = A + \frac{(1+R_1)A + T - C_1 - \tau_k R_1 A - \tau_a A}{1+r}; \quad \mathcal{A}^h = A + \frac{(1+R_1)A + T - C_1 - \tau_a A}{1+r}. \quad (\text{A10})$$

Indirect utility V is continuous at $R_2 = \bar{R}(R_1)$ because the investor is indifferent between selling and keeping the asset. The Euler equation (6) then implies that $C_1(\cdot)$ is continuous at the cutoff. Therefore, the jump in each tax base can be computed by evaluating $\mathcal{J}^s - \mathcal{J}^h$ and $\mathcal{A}^s - \mathcal{A}^h$ at $R_2 = \bar{R}(R_1)$. Using the optimal reservation return in (8) evaluated with $\tau_R = 0$, we obtain

$$\mathcal{J}^s - \mathcal{J}^h = \frac{[r + r(2 - \tau_k)R_1 - \bar{R}(R_1)(1 + R_1)] A}{1 + r} = \frac{\left(r - \frac{\tau_k \tau_a}{1 - \tau_k}\right) R_1 A}{1 + r},$$

which proves (19). The same steps deliver (20). $\square$

Proposition A3 (Optimal taxes). *Optimal taxes $\{\tau_k, \tau_a, T\}$ satisfy*

$$1 - \bar{g}^J \leq \frac{\tau_k}{1 - \tau_k} \left(\bar{\varepsilon}_{J, 1 - \tau_k}^J + \bar{\xi}_{J, 1 - \tau_k}^J \right) + \frac{\tau_a}{1 - \tau_k} \frac{\bar{\mathcal{A}}}{\bar{J}} \left(\bar{\varepsilon}_{\mathcal{A}, 1 - \tau_k}^A + \bar{\xi}_{\mathcal{A}, 1 - \tau_k}^A \right), \quad (\text{A11})$$

$$1 - \bar{g}^A \leq \frac{\tau_a}{1 - \tau_a} \left(\bar{\varepsilon}_{\mathcal{A}, 1 - \tau_a}^A + \bar{\xi}_{\mathcal{A}, 1 - \tau_a}^A \right) + \frac{\tau_k}{1 - \tau_a} \frac{\bar{J}}{\bar{\mathcal{A}}} \left(\bar{\varepsilon}_{J, 1 - \tau_a}^J + \bar{\xi}_{J, 1 - \tau_a}^J \right), \quad (\text{A12})$$

$$\bar{g} \leq 1 - \mathbb{E} \left[\tau_k \frac{\partial \mathcal{J}}{\partial T} + \tau_a \frac{\partial \mathcal{A}}{\partial T} \right]. \quad (\text{A13})$$

Here, $\bar{x} \equiv \mathbb{E}[x(\cdot)]$ denotes the population average of $x(\cdot)$, and $\bar{x}^y \equiv \mathbb{E}[x(\cdot) \mathcal{Y}(\cdot)] / \mathbb{E}[\mathcal{Y}(\cdot)]$ denotes the average weighted by tax base $\mathcal{Y} \in \{J, A\}$.

PROOF. The government's Lagrangian can be written as

$$\begin{aligned} \mathcal{L} &= \int \left[\frac{\alpha V(\cdot)}{\eta} + \tau_k \mathcal{J}(\cdot) + \tau_a \mathcal{A}(\cdot) - T - G \right] dF(A, R_1, R_2), \\ &= \int \int_{\underline{R}}^{\bar{R}(R_1)} \left(\frac{\alpha V(\cdot)}{\eta} + \tau_k \mathcal{J}(\cdot) + \tau_a \mathcal{A}(\cdot) \right) dF^{R_2}(R_2 | A, R_1) dF(A, R_1) \\ &\quad + \int \int_{\bar{R}(R_1)}^{\bar{R}} \left(\frac{\alpha V(\cdot)}{\eta} + \tau_k \mathcal{J}(\cdot) + \tau_a \mathcal{A}(\cdot) \right) dF^{R_2}(R_2 | A, R_1) dF(A, R_1) - T - G, \end{aligned} \quad (\text{A14})$$

where η is the multiplier on the government budget constraint (16), $F^{R_2}(\cdot | A, R_1)$ denotes the conditional distribution of R_2 given (A, R_1) , and $f^{R_2}(\cdot | A, R_1)$ denotes the corresponding conditional density. The population is split around the reservation return $\bar{R}(R_1)$ defined in (8).

The first-order conditions for τ_k , τ_a , and T are:

$$0 \geq \mathbb{E} \left[\frac{\alpha}{\eta} \frac{\partial V}{\partial \tau_k} + \tau_k \frac{\partial J}{\partial \tau_k} + \tau_a \frac{\partial \mathcal{A}}{\partial \tau_k} + J \right] + \mathbb{E} \left[(\tau_k \Delta J + \tau_a \Delta \mathcal{A}) f^{R_2} (\bar{R}(R_1)|A, R_1) \frac{\partial \bar{R}(R_1)}{\partial \tau_k} \right], \quad (\text{A15})$$

$$0 \geq \mathbb{E} \left[\frac{\alpha}{\eta} \frac{\partial V}{\partial \tau_a} + \tau_k \frac{\partial J}{\partial \tau_a} + \tau_a \frac{\partial \mathcal{A}}{\partial \tau_a} + \mathcal{A} \right] + \mathbb{E} \left[(\tau_k \Delta J + \tau_a \Delta \mathcal{A}) f^{R_2} (\bar{R}(R_1)|A, R_1) \frac{\partial \bar{R}(R_1)}{\partial \tau_a} \right], \quad (\text{A16})$$

$$0 \geq \mathbb{E} \left[\frac{\alpha}{\eta} \frac{\partial V}{\partial T} + \tau_k \frac{\partial J}{\partial T} + \tau_a \frac{\partial \mathcal{A}}{\partial T} - 1 \right]. \quad (\text{A17})$$

The inequalities bind when the corresponding lower-bound constraints do not bind.

Start from the first-order conditions (A15)-(A17). The envelope conditions imply that a marginal increase in τ_k or τ_a reduces private resources in proportion to the relevant tax base, while a marginal increase in T raises period-1 resources one for one. Using this and substituting the welfare weights in (22), the two tax-rate first-order conditions become:

$$\begin{aligned} \mathbb{E} [(g-1)J] &\geq \mathbb{E} \left[\tau_k \frac{\partial J}{\partial \tau_k} + \tau_a \frac{\partial \mathcal{A}}{\partial \tau_k} \right] + \mathbb{E} \left[\frac{\partial \bar{R}}{\partial \tau_k} (\tau_k \Delta J + \tau_a \Delta \mathcal{A}) f^{R_2} \right], \\ \mathbb{E} [(g-1)\mathcal{A}] &\geq \mathbb{E} \left[\tau_k \frac{\partial J}{\partial \tau_a} + \tau_a \frac{\partial \mathcal{A}}{\partial \tau_a} \right] + \mathbb{E} \left[\frac{\partial \bar{R}}{\partial \tau_a} (\tau_k \Delta J + \tau_a \Delta \mathcal{A}) f^{R_2} \right]. \end{aligned}$$

Dividing the first inequality by $-\bar{J}$ and the second by $-\bar{\mathcal{A}}$, and substituting the definitions of $\varepsilon_{y,1-\tau}$ and $\xi_{y,1-\tau}$, gives (25) and (26). The condition for T follows directly from (A17) and (22). $\square$

B.1. Marginal excess burdens and the determinants of optimal taxes

We now tackle the optimal mix of capital income and wealth taxes that satisfies the conditions in Proposition 3 summarized in (28).

The optimal tax mix therefore equates the marginal burden of the two instruments after adjusting for the redistributive value of the tax bases. The terms in (28) can be decomposed into three channels: an intertemporal tax-base channel and a portfolio-correction channel that operate through the marginal burden responses, and a redistributive channel that depends on the welfare weights.

The intertemporal tax-base channel. This channel operates only through changes in investors' savings choices. It corresponds to the case discussed in B.2 and provides a necessary condition for the presence of capital income taxes.

The portfolio-correction channel. When $\tau_k > 0$ and τ_a is low, (8) implies that the reservation return lies below the market return r , generating lock-in. The portfolio

elasticity terms in (25) and (26) capture the fiscal consequences of moving this cutoff:

$$\begin{aligned} \text{MEB}_k^{\text{Portfolio}} &= \frac{\tau_k}{1-\tau_k} \bar{\xi}_{J,1-\tau_k}^J + \frac{\tau_a}{1-\tau_k} \frac{\bar{\mathcal{A}}}{\bar{J}} \bar{\xi}_{\mathcal{A},1-\tau_k}^{\mathcal{A}}, \\ \text{MEB}_a^{\text{Portfolio}} &= \frac{\tau_a}{1-\tau_a} \bar{\xi}_{\mathcal{A},1-\tau_a}^{\mathcal{A}} + \frac{\tau_k}{1-\tau_a} \frac{\bar{J}}{\bar{\mathcal{A}}} \bar{\xi}_{J,1-\tau_a}^J. \end{aligned} \quad (\text{A18})$$

The signs of these terms follow from the tax-base jumps in Lemma 1, with negative cross-base terms and positive own-base terms, as discussed in Section 4.4. The negative cross-base term is the fiscal gain from using a wealth tax to unlock capital income. Similar to the discussion on condition (29), this fiscal gain is more important when the capital income tax rate is high and when the capital income tax base is large relative to the wealth tax base.

This logic shows why it is not (in general) optimal to set wealth taxes to restore efficiency by setting $\tau_a = \tau_a^*(\tau_k, 0)$ as in Corollary 1. At $\tau_a^*(\tau_k, 0)$, the revenue jump from moving a marginal investor across the realization cutoff is zero by (21). At that point, the portfolio-correction benefit of raising τ_a has been exhausted. If capital income taxes dominate wealth taxes on the intertemporal margin, as in (A20), then the remaining force pushes the government toward a lower wealth tax and a higher capital income tax. Conversely, when the reservation return is above r , there is capital lock-out and further increasing the wealth tax would worsen portfolio efficiency.

Putting the intertemporal and portfolio correction channels together, the optimal wealth tax satisfies $\tau_a \in [0, \tau_a^*(\tau_k, 0))$. An interior solution with $\tau_k, \tau_a > 0$ requires $\text{MEB}_k > \text{MEB}_a$ when $(\tau_k, \tau_a) = (\tau_k|_{\tau_a=0}^*, 0)$, so that wealth taxes are favored when $\tau_a = 0$ and capital income taxes are set to their optimal value under no wealth taxes, $\tau_k|_{\tau_a=0}^*$. The required condition, aside from distributional motives, is

$$\frac{\bar{\xi}_{J,1-\tau_k}^J + \bar{\xi}_{J,1-\tau_k}^J}{\left(1 - \tau_k|_{\tau_a=0}^*\right) \bar{J}} > \frac{\bar{\xi}_{J,1-\tau_a}^J + \bar{\xi}_{J,1-\tau_a}^J}{\bar{\mathcal{A}}}. \quad (\text{A19})$$

The redistributive channel. The final term in (28) captures redistribution across tax bases. If $\bar{g}^J < \bar{g}^{\mathcal{A}}$, capital income is concentrated in investors with lower social welfare weights than investors who hold wealth. The redistribution channel then favors using the capital income tax, and the government is willing to accept a higher behavioral distortion from τ_k . If instead $\bar{g}^{\mathcal{A}} < \bar{g}^J$, wealth is concentrated in investors with low social welfare weights, and the redistribution channel favors the wealth tax. This would be needed for the optimal wealth tax to exceed the level justified by the intertemporal and portfolio-efficiency channels. For example, the wealth tax can exceed the portfolio-efficient rate $\tau_a^*(\tau_k, 0)$ if the redistributive value of taxing wealth is large enough to compensate for the resulting lock-out.

B.2. A necessary condition for capital income taxation

Finally, we derive a simple necessary condition for the desirability of capital income taxes. These taxes distort the portfolio choice of investors generating lock-in and thus the justification for their introduction must come from their advantage in raising revenue with a lower distortion of investors' intertemporal savings choice.

We can better see this if we abstract for a moment from other channels and look at a hypothetical benchmark in which portfolio choices are efficient and do not respond to taxes.¹⁹ In this case, capital income and wealth taxes operate by raising revenue while reducing the effective return ($\tilde{r} = (1 - \tau_k) r - \tau_a$). The question for the government becomes: which tax instrument yields more revenue for a given reduction in $\tilde{r}$?

The answer turns out to be to tax the largest base, so that a lower rate is required per unit of revenue. This hinges on whether excess returns in the first period (relative to the market rate) are larger than the wealth tax base that covers the wealth in both periods.

Lemma A2 (Necessary condition for capital income taxation). *In the absence of portfolio distortions, a marginal increase in the capital income tax is desirable on efficiency grounds when*

$$\text{MEB}_k^B < \text{MEB}_a^B \iff \bar{\mathcal{J}}_0 > r\bar{\mathcal{A}}_0, \quad (\text{A20})$$

where $\bar{\mathcal{J}}_0$ and $\bar{\mathcal{A}}_0$ denote the tax bases evaluated under the efficient portfolio allocation.

PROOF. Both τ_k and τ_a operate on the savings choice by reducing the effective return ($\tilde{r} = (1 - \tau_k) r - \tau_a$). Their compensated tax-base responses are then linked by:

$$\begin{aligned} \left. \frac{\partial \mathcal{J}}{\partial \tau_k} \right|_{V=V_0} &= -\frac{r^2}{1+r} \left. \frac{\partial B}{\partial \tilde{r}} \right|_{V=V_0}, & \left. \frac{\partial \mathcal{J}}{\partial \tau_a} \right|_{V=V_0} &= -\frac{r}{1+r} \left. \frac{\partial B}{\partial \tilde{r}} \right|_{V=V_0}, \\ \left. \frac{\partial \mathcal{A}}{\partial \tau_k} \right|_{V=V_0} &= -\frac{r}{1+r} \left. \frac{\partial B}{\partial \tilde{r}} \right|_{V=V_0}, & \left. \frac{\partial \mathcal{A}}{\partial \tau_a} \right|_{V=V_0} &= -\frac{1}{1+r} \left. \frac{\partial B}{\partial \tilde{r}} \right|_{V=V_0}. \end{aligned} \quad (\text{A21})$$

These expressions follow because a marginal increase in τ_k reduces $\tilde{r}$ by r , while a marginal increase in τ_a reduces $\tilde{r}$ by one.

Using the definition of the compensated elasticities in (23) and the definition of the excess burdens in (25) and (26), the intertemporal components are

$$\text{MEB}_k^B = -\frac{1}{\bar{\mathcal{J}}_0} \mathbb{E} \left[\tau_k \left. \frac{\partial \mathcal{J}}{\partial \tau_k} \right|_{V=V_0} + \tau_a \left. \frac{\partial \mathcal{A}}{\partial \tau_k} \right|_{V=V_0} \right], \quad \text{MEB}_a^B = -\frac{1}{\bar{\mathcal{A}}_0} \mathbb{E} \left[\tau_k \left. \frac{\partial \mathcal{J}}{\partial \tau_a} \right|_{V=V_0} + \tau_a \left. \frac{\partial \mathcal{A}}{\partial \tau_a} \right|_{V=V_0} \right], \quad (\text{A22})$$

where $\bar{\mathcal{J}}_0$ and $\bar{\mathcal{A}}_0$ denote the tax bases evaluated under the efficient portfolio allocation.

¹⁹Formally, set the cutoff at the efficient threshold, $\bar{R}(R_1) = r$, and suppress the portfolio elasticities, $\xi_{y,1-\tau} = 0$. The only behavioral margin is then savings in the risk-free asset B .

Substituting (A21) into (A22) gives

$$\text{MEB}_k^B = \frac{\tau_k r^2 + \tau_a r}{(1+r)\bar{J}_0} \mathbb{E} \left[\left. \frac{\partial B}{\partial \tilde{r}} \right|_{V=V_0} \right], \quad \text{MEB}_a^B = \frac{\tau_k r + \tau_a}{(1+r)\bar{A}_0} \mathbb{E} \left[\left. \frac{\partial B}{\partial \tilde{r}} \right|_{V=V_0} \right]. \quad (\text{A23})$$

Assuming a positive aggregate compensated savings response and $\tau_k r + \tau_a > 0$, the capital income tax has the lower intertemporal marginal excess burden if and only if (A20).

Equivalently, after substituting the tax bases, this condition can be written as

$$\begin{aligned} \int \left(\int_{\underline{R}}^r A(R_1 - r) dF^{R_2}(R_2|A, R_1) + \int_r^{\bar{R}} \frac{A[(R_1 - r) + (R_2 - r)(1 + R_1)]}{1 + r} dF^{R_2}(R_2|A, R_1) \right) dF(A, R_1) \\ > \int \int_r^{\bar{R}} \frac{r^2 A}{1 + r} dF^{R_2}(R_2|A, R_1) dF(A, R_1). \end{aligned} \quad (\text{A24})$$

The left-hand side collects excess returns relative to the normal return on wealth. The right-hand side is the revenue-timing advantage of the wealth tax as it is collected in both periods, whereas capital income taxes collect only when returns are realized. $\square$

An important implication of our previous results in Proposition 1 is that if (A20) is violated, the capital income tax is not desirable on efficiency grounds: wealth taxes distort savings less per unit of revenue than capital income taxes, and do not generate portfolio inefficiencies on their own. Provided redistributive motives do not favor taxing capital-income earners over wealth holders, $\bar{g}^J \geq \bar{g}^A$, this is enough to imply τ_k should equal zero, so that (A20) becomes a necessary condition for a positive capital income tax.

Appendix C. Optimal taxes with unrealized capital gains

This appendix extends the analysis of Section 4 by adding the tax on unrealized capital gains, τ_R . We provide general formulas for optimal taxes and a sufficient condition for having $\tau_R > 0$ as part of the optimal tax mix. The sufficient condition is almost always satisfied in our two-period model but it would not be in more general settings.

C.1. Tax bases and jumps at the realization cutoff

When τ_R is available, the government budget constraint becomes

$$G + T \leq \int [\tau_k \mathcal{J}(A, R_1, R_2) + \tau_a \mathcal{A}(A, R_1, R_2) + \tau_R \mathcal{Z}(A, R_1, R_2)] dF(A, R_1, R_2), \quad (\text{A25})$$

where $\mathcal{J}(\cdot)$ and $\mathcal{A}(\cdot)$ are defined in (17) and (18), and

$$\mathcal{Z}(A, R_1, R_2) \equiv (1 - \phi^*(A, R_1, R_2)) R_1 A \quad (\text{A26})$$

is the tax base for unrealized capital gains. The reservation return is the one in (8). We proceed by generalizing Lemma 1 for the jumps in tax bases when $\tau_R > 0$.

Lemma A3 (Tax-base jumps with unrealized gains taxes). *Assume $\tau_k, \tau_a, \tau_R \geq 0$, $\tau_k < 1$, and $1 + \tilde{r} > 0$. At the reservation return $R_2 = \bar{R}(R_1)$, the jumps in tax bases from moving a marginal investor from holding to selling are*

$$\Delta \mathcal{J}(A, R_1) \equiv \lim_{R_2 \rightarrow \bar{R}(R_1)} \mathcal{J}(A, R_1, R_2) - \lim_{R_2 \rightarrow^+ \bar{R}(R_1)} \mathcal{J}(A, R_1, R_2) = \frac{(1 - \tau_k)r - \tau_k \tau_a - (1 - \tau_a)\tau_R}{1 - \tau_k} \frac{R_1 A}{1 + r}, \quad (\text{A27})$$

$$\Delta \mathcal{A}(A, R_1) \equiv \lim_{R_2 \rightarrow \bar{R}(R_1)} \mathcal{A}(A, R_1, R_2) - \lim_{R_2 \rightarrow^+ \bar{R}(R_1)} \mathcal{A}(A, R_1, R_2) = -(\tau_k - \tau_R) \frac{R_1 A}{1 + r}, \quad (\text{A28})$$

$$\Delta \mathcal{Z}(A, R_1) \equiv \lim_{R_2 \rightarrow \bar{R}(R_1)} \mathcal{Z}(A, R_1, R_2) - \lim_{R_2 \rightarrow^+ \bar{R}(R_1)} \mathcal{Z}(A, R_1, R_2) = -R_1 A. \quad (\text{A29})$$

PROOF. Let $C_1(\cdot)$ be the optimal first-period consumption of the marginal investor. If the marginal investor sells, $\phi = 1$, the period-1 budget gives

$$B^s = (1 + R_1)A + T - C_1 - \tau_k R_1 A - \tau_a A.$$

If the marginal investor holds, $\phi = 0$, the period-1 budget gives

$$B^h = T - C_1 - \tau_a A - \tau_R R_1 A.$$

Substituting B^s and B^h into the tax bases gives

$$\begin{aligned} \mathcal{J}^s &= R_1 A + r \frac{(1+R_1)A + T - C_1 - \tau_k R_1 A - \tau_a A}{1+r}, \\ \mathcal{J}^h &= \frac{[R_1 + R_2(1+R_1)] A + r(T - C_1 - \tau_a A - \tau_R R_1 A)}{1+r}, \\ \mathcal{A}^s &= A + \frac{(1+R_1)A + T - C_1 - \tau_k R_1 A - \tau_a A}{1+r}, \\ \mathcal{A}^h &= A + \frac{(1+R_1)A + T - C_1 - \tau_a A - \tau_R R_1 A}{1+r}, \\ \mathcal{Z}^s &= 0, \quad \mathcal{Z}^h = R_1 A. \end{aligned}$$

Since the investor is indifferent between selling and holding at $R_2 = \bar{R}(R_1)$, indirect utility is continuous at the cutoff. The Euler equation (6) then implies that $C_1(\cdot)$ is continuous at the cutoff. Evaluating $\mathcal{J}^s - \mathcal{J}^h$, $\mathcal{A}^s - \mathcal{A}^h$, and $\mathcal{Z}^s - \mathcal{Z}^h$ at $R_2 = \bar{R}(R_1)$ gives the stated jumps. $\square$

The total revenue jump from moving a marginal investor from holding to selling is

$$\Delta \mathcal{R}(A, R_1) \equiv \tau_k \Delta \mathcal{J}(A, R_1) + \tau_a \Delta \mathcal{A}(A, R_1) + \tau_R \Delta \mathcal{Z}(A, R_1). \quad (\text{A30})$$

When $\tau_R = 0$, (A27)–(A30) collapse to Lemma 1 and (21).

C.2. Sufficient statistics for optimal taxes

For any tax base $\mathcal{Y} \in \{\mathcal{J}, \mathcal{A}, \mathcal{Z}\}$ and tax rate $\tau \in \{\tau_k, \tau_a, \tau_R\}$, define the intertemporal tax-base elasticity $\varepsilon_{\mathcal{Y}, 1-\tau}(A, R_1, R_2)$ and the portfolio elasticity $\xi_{\mathcal{Y}, 1-\tau}(A, R_1)$ as before.

Notice that τ_R does not change the effective market return $\tilde{r}$. Thus, the intertemporal components with respect to τ_R are zero away from the portfolio margin in our model. We keep the $\varepsilon_{\mathcal{Y}, 1-\tau_R}$ terms below so that the expression has the same sufficient-statistics structure as the optimal-tax conditions in the main text.

Proposition A4 (Optimal taxes with unrealized gains). *When the government chooses $\{\tau_k, \tau_a, \tau_R, T\}$, with $\tau_R < 1$ so that the net-of-tax rate $1 - \tau_R$ is positive, optimal taxes satisfy*

$$\begin{aligned} 1 - \bar{g}^{\mathcal{J}} &\leq \frac{\tau_k}{1 - \tau_k} \left(\bar{\varepsilon}_{\mathcal{J}, 1-\tau_k}^{\mathcal{J}} + \bar{\xi}_{\mathcal{J}, 1-\tau_k}^{\mathcal{J}} \right) + \frac{\tau_a}{1 - \tau_k} \frac{\bar{A}}{\bar{J}} \left(\bar{\varepsilon}_{\mathcal{A}, 1-\tau_k}^{\mathcal{A}} + \bar{\xi}_{\mathcal{A}, 1-\tau_k}^{\mathcal{A}} \right) \\ &\quad + \frac{\tau_R}{1 - \tau_k} \frac{\bar{Z}}{\bar{J}} \left(\bar{\varepsilon}_{\mathcal{Z}, 1-\tau_k}^{\mathcal{Z}} + \bar{\xi}_{\mathcal{Z}, 1-\tau_k}^{\mathcal{Z}} \right), \end{aligned} \quad (\text{A31})$$

$$\begin{aligned} 1 - \bar{g}^{\mathcal{A}} &\leq \frac{\tau_a}{1 - \tau_a} \left(\bar{\varepsilon}_{\mathcal{A}, 1-\tau_a}^{\mathcal{A}} + \bar{\xi}_{\mathcal{A}, 1-\tau_a}^{\mathcal{A}} \right) + \frac{\tau_k}{1 - \tau_a} \frac{\bar{J}}{\bar{A}} \left(\bar{\varepsilon}_{\mathcal{J}, 1-\tau_a}^{\mathcal{J}} + \bar{\xi}_{\mathcal{J}, 1-\tau_a}^{\mathcal{J}} \right) \\ &\quad + \frac{\tau_R}{1 - \tau_a} \frac{\bar{Z}}{\bar{A}} \left(\bar{\varepsilon}_{\mathcal{Z}, 1-\tau_a}^{\mathcal{Z}} + \bar{\xi}_{\mathcal{Z}, 1-\tau_a}^{\mathcal{Z}} \right), \end{aligned} \quad (\text{A32})$$

$$1 - \bar{g}^{\mathcal{Z}} \leq \frac{\tau_R}{1 - \tau_R} \left(\bar{\varepsilon}_{\mathcal{Z}, 1-\tau_R}^{\mathcal{Z}} + \bar{\xi}_{\mathcal{Z}, 1-\tau_R}^{\mathcal{Z}} \right) + \frac{\tau_k}{1 - \tau_R} \frac{\bar{J}}{\bar{Z}} \left(\bar{\varepsilon}_{\mathcal{J}, 1-\tau_R}^{\mathcal{J}} + \bar{\xi}_{\mathcal{J}, 1-\tau_R}^{\mathcal{J}} \right)$$

$$+ \frac{\tau_a}{1 - \tau_R} \frac{\bar{\mathcal{A}}}{\bar{\mathcal{Z}}} \left(\bar{\varepsilon}_{\mathcal{A}, 1 - \tau_R}^{\mathcal{A}} + \bar{\xi}_{\mathcal{A}, 1 - \tau_R}^{\mathcal{A}} \right), \quad (\text{A33})$$

$$1 - \bar{g} \geq \mathbb{E} \left[\tau_k \frac{\partial \mathcal{J}}{\partial T} + \tau_a \frac{\partial \mathcal{A}}{\partial T} \right]. \quad (\text{A34})$$

The inequalities bind when the corresponding lower-bound constraints do not bind.

PROOF. The proof follows the same steps as the proof of Proposition 3. The Lagrangian is

$$\mathcal{L} = \int \left[\frac{\alpha^i V(\cdot)}{\eta} + \tau_k \mathcal{J}(\cdot) + \tau_a \mathcal{A}(\cdot) + \tau_R \mathcal{Z}(\cdot) - T - G \right] dF(A, R_1, R_2),$$

with the population split around the cutoff $\bar{R}(R_1)$. The first-order condition for τ_R is

$$0 \geq \mathbb{E} \left[\frac{\alpha^i}{\eta} \frac{\partial V}{\partial \tau_R} + \tau_k \frac{\partial \mathcal{J}}{\partial \tau_R} + \tau_a \frac{\partial \mathcal{A}}{\partial \tau_R} + \tau_R \frac{\partial \mathcal{Z}}{\partial \tau_R} + \mathcal{Z} \right] + \mathbb{E} \left[\Delta \mathcal{R}(A, R_1) f^{R_2}(\bar{R}(R_1)|A, R_1) \frac{\partial \bar{R}(R_1)}{\partial \tau_R} \right]. \quad (\text{A35})$$

The envelope condition gives

$$\frac{\alpha^i}{\eta} \frac{\partial V}{\partial \tau_R} = - \frac{\alpha^i U_{C_1}}{\eta} \mathcal{Z}.$$

Substituting the definition of g^i gives

$$\mathbb{E} [(g - 1)\mathcal{Z}] \geq \mathbb{E} \left[\tau_k \frac{\partial \mathcal{J}}{\partial \tau_R} + \tau_a \frac{\partial \mathcal{A}}{\partial \tau_R} + \tau_R \frac{\partial \mathcal{Z}}{\partial \tau_R} \right] + \mathbb{E} \left[\Delta \mathcal{R}(A, R_1) f^{R_2}(\bar{R}(R_1)|A, R_1) \frac{\partial \bar{R}(R_1)}{\partial \tau_R} \right]. \quad (\text{A36})$$

Dividing by $-\bar{\mathcal{Z}}$ and substituting the intertemporal and portfolio elasticities gives (A33). The conditions for τ_k and τ_a are obtained by the same argument, using $\mathcal{J}$ and $\mathcal{A}$ as the direct tax bases, respectively. The condition for T follows from the first-order condition for transfers. $\square$

C.3. A sufficient condition for unrealized-gains taxes

We now provide a sufficient condition for introducing taxes on unrealized gains. We start by evaluating (A33) with a positive capital income tax, $\tau_k > 0$, but no wealth tax or unrealized-gains tax, $\tau_a = \tau_R = 0$. Let a subscript 0 denote objects evaluated at this point. Assume $\bar{\mathcal{Z}}_0 > 0$. A positive unrealized-gains tax is locally desirable if

$$1 - \bar{g}_0^{\mathcal{Z}} > \tau_k \frac{\bar{\mathcal{J}}_0}{\bar{\mathcal{Z}}_0} \left(\bar{\varepsilon}_{\mathcal{J}, 1 - \tau_R, 0}^{\mathcal{J}} + \bar{\xi}_{\mathcal{J}, 1 - \tau_R, 0}^{\mathcal{J}} \right), \quad (\text{A37})$$

As for wealth taxes, the relevant statistics are the cross-elasticities of the capital income tax base with respect to the unrealized capital gains tax.

In our two-period model, the unrealized-gains tax does not change the effective market return $\tilde{r}$. Therefore, $\bar{\varepsilon}_{J,1-\tau_R,0}^J = 0$. The condition in (A37) then becomes

$$1 - \bar{g}_0^Z > \tau_k \frac{\bar{J}_0}{\bar{Z}_0} \bar{\varepsilon}_{J,1-\tau_R,0}^J. \quad (\text{A38})$$

The sign of the portfolio term is determined by the way τ_R moves the reservation return:

$$\left. \frac{\partial \bar{R}(R_1)}{\partial \tau_R} \right|_0 = \frac{1 + (1 - \tau_k) r}{1 - \tau_k} \frac{R_1}{1 + R_1} > 0. \quad (\text{A39})$$

At the same point, the capital income tax base jumps by

$$\Delta J_0(A, R_1) = r \frac{R_1 A}{1 + r} > 0. \quad (\text{A40})$$

Hence, if there is positive density at the realization cutoff for investors with $R_1 A > 0$, raising τ_R moves marginal investors from holding to selling and expands the capital income tax base. This corresponds to the lock-out effect of unrealized capital gains taxes. By the definition of the elasticity, this implies $\bar{\varepsilon}_{J,1-\tau_R,0}^J < 0$. Combining this with (A38) yields a simple sufficient condition:

$$\bar{g}_0^Z \leq 1 \quad \text{and} \quad \bar{\varepsilon}_{J,1-\tau_R,0}^J < 0 \quad \implies \quad \tau_R^{\text{opt}} > 0. \quad (\text{A41})$$

Thus, absent a redistributive reason to avoid taxing holders of unrealized gains, an unrealized-gains tax is locally desirable whenever a positive capital income tax leaves residual lock-in. As in Corollary 2, a tax is attractive when it expands another tax base by correcting portfolio distortions. The difference is that, in our setting, τ_R targets the realization margin without affecting the intertemporal margin (a small τ_R raises $\bar{R}(R_1)$ directly and leaves $\tilde{r}$ unchanged). So, the sufficient condition for $\tau_R > 0$ contains the portfolio cross-base term but no offsetting intertemporal term.

However, this result is special to our two-period setup and does not generalize. It should not be interpreted as an argument for unrealized capital gains taxation. In our setup, all capital gains are realized in period 2, so the tax τ_R applies only to the period-1 stock of unrealized gains and does not change the return on the risk-free asset. With more periods, τ_R would apply repeatedly to returns that remain unrealized over time. Then, if an investment is rolled over and its accrued return is taxed before final realization, the effective market return would be $\tilde{r} = (1 - \tau_k - \tau_R) r - \tau_a$. This makes it plain that τ_R is not, in practice, a pure realization-margin instrument. Instead, τ_R also lowers the intertemporal return and generates a savings response, $\bar{\varepsilon}_{J,1-\tau_R,0}^J > 0$, so condition (A37) applies in full. An unrealized-gains tax is therefore not necessarily desirable, even when there is residual lock-in, because its corrective effect on realization could be offset by the distortion it creates on the market return.

Appendix D. Alternative wealth-tax base

D.1. Market value wealth taxes and the double taxation of capital gains

We analyze here the market-value wealth tax, τ_m , described in section 5. We start from equation (31) and denote this second component by $\tilde{\tau}_R$ to distinguish it from the tax on unrealized gains, τ_R , studied above. We keep τ_a and $\tilde{\tau}_R$ separate in the derivations below and then impose $\tau_a = \tilde{\tau}_R = \tau_m$ to recover the market-value wealth tax.

Distinguishing between τ_a and $\tilde{\tau}_R$ helps us see how the market-value wealth tax affects the effective returns in the second period. Second period market-value wealth taxes, $\tau_m W_2^i$, bundle taxes on the historical value of wealth, $\tau_a \left((1 - \phi^i) (1 + R_1^i) A^i + B^i \right)$, and on accrued capital gains, $\tilde{\tau}_R \left(R_2^i (1 - \phi^i) (1 + R_1^i) A^i + r B^i \right)$. The effective returns are now

$$\tilde{R}_{2,m}^i \equiv (1 - \tau_k - \tilde{\tau}_R) R_2^i - \tau_a, \quad \tilde{r}_m \equiv (1 - \tau_k - \tilde{\tau}_R) r - \tau_a. \quad (\text{A42})$$

The intertemporal budget constraint is now determined by the new effective returns in (A42) and includes the taxation of first period returns via market-value wealth taxes on the wealth endowment of investors:

$$\underbrace{\left(\frac{1 + \tilde{R}_{2,m}^i}{1 + \tilde{r}_m} (1 - \phi^i) + \phi^i \right) (1 + R_1^i) A^i + T}_{\text{NPV Assets}} \quad (\text{A43})$$

$$= C_1^i + \frac{C_2^i}{1 + \tilde{r}_m} + \underbrace{\tau_a A^i + \tilde{\tau}_R R_1^i A^i}_{\text{Tax on market value of endowment}} + \underbrace{\left(\tau_k \phi^i + \frac{\tau_k}{1 + \tilde{r}_m} (1 - \phi^i) \right) R_1^i A^i}_{\text{NPV tax on first-period gains}}.$$

The portfolio choice of investors determining the realization of their capital gains is characterized by a reservation return paralleling (8),

$$\bar{R}_m(R_1^i) \equiv r - \tau_k r \frac{R_1^i}{1 + R_1^i} + \frac{\tau_k \tau_a}{1 - \tau_k - \tilde{\tau}_R} \frac{R_1^i}{1 + R_1^i}, \quad (\text{A44})$$

where the middle term captures the lock-in incentives of capital income taxes and the third term the lock-out incentives of wealth taxes (this term grows with τ_m by simultaneously increasing the numerator through τ_a and decreasing the denominator through $\tilde{\tau}_R$). The lock-out effect of wealth taxes is once again only present under capital income taxes. The reason is that $\tilde{\tau}_R$ is paid regardless of realization, so it does not change the relative tax cost of realizing or delaying the first-period gain. It affects the

reservation return only indirectly, through the effective discount rate in (A42).

We can now explore how a market-value wealth tax unlocks capital.

Proposition A5 (Market-value wealth taxes unlock capital). *Portfolio choice is efficient for all investors if and only if there are no capital income taxes, $\tau_k = 0$, or if the effective market return is zero, $\tilde{r}_m = (1 - \tau_k - \tilde{\tau}_R) r - \tau_a = 0$, which happens under a market-value tax ($\tau_m = \tau_a = \tilde{\tau}_R$) equal to*

$$\tau_m^*(\tau_k) = \frac{(1 - \tau_k) r}{1 + r}. \quad (\text{A45})$$

PROOF. Assume $\tau_k + \tilde{\tau}_R < 1$. From (A44), portfolio choice is efficient for all investors if and only if the coefficient multiplying $R_1^i / (1 + R_1^i)$ equals zero: $\tau_k r - \frac{\tau_k \tau_a}{1 - \tau_k - \tilde{\tau}_R} = 0$. Since $\tau_k > 0$, this condition is equivalent to $\tau_a = (1 - \tau_k - \tilde{\tau}_R) r$, which implies zero effective market return from (A42). Substituting $\tau_a = \tilde{\tau}_R = \tau$ gives (A45). The same result follows if $\tau_k + \tilde{\tau}_R = 1$ starting from (A43) to avoid indeterminacies. $\square$

A market-value wealth tax can unlock capital gains by reducing the effective market return to zero, eliminating the tax advantage of delaying capital gains. This is the same result as in case (a) of Corollary 1 when there are no unrealized capital gains taxes, $\tau_R = 0$. This reflects the fact that the tax on accrued capital gains, $\tilde{\tau}_R$, does not generate lock-out incentives by itself, unlike τ_R . Instead, $\tilde{\tau}_R$ lowers the required tax on historical wealth because it already lowers the effective market return, unlike τ_R .

D.2. Updating wealth tax base and lock-in forces

The historical wealth tax in the main text assumes the tax falls on the value of assets carried into each period. An alternative is to tax the original principal without updating its value. This taxes

$$(1 - \phi^i) A^i + B^i. \quad (\text{A46})$$

This alternative is useful because it separates the role of updating the historical basis from the role of taxing the market value of wealth studied in Section 5.

Under this alternative base, the period-2 budget becomes

$$\left((1 + (1 - \tau_k) R_2^i) (1 + R_1^i) - \tau_a \right) (1 - \phi^i) A^i + (1 + \tilde{r}) B^i = C_2^i + \tau_k R_1^i (1 - \phi^i) A^i. \quad (\text{A47})$$

Combining this budget with the period-1 budget gives the intertemporal budget

$$\left[\frac{\left(1 + (1 - \tau_k) R_2^i\right) \left(1 + R_1^i\right) - \tau_a}{1 + \tilde{r}} \left(1 - \phi^i\right) + \phi^i \left(1 + R_1^i\right) \right] A^i + T = C_1^i + \frac{C_2^i}{1 + \tilde{r}} + \tau_a A^i + \left[\tau_k \phi^i + \left(\tau_R + \frac{\tau_k}{1 + \tilde{r}} \right) \left(1 - \phi^i\right) \right] R_1^i A^i. \quad (\text{A48})$$

Relative to (4), the first term discounts the after-tax payoff on the original asset base rather than the after-tax return on the period-1 value of unsold assets.

The portfolio condition changes because delaying realization no longer increases the period-2 wealth-tax base by the first-period capital gain. In fact, taxing only the original value of the principal gives the investor an incentive to retain capital gains so as not to increase the value of taxable wealth. We see this clearly in the reservation return,

$$\bar{R}_A \left(R_1^i \right) \equiv r - \left(\tau_a + \tau_k r - \frac{1 + \tilde{r}}{1 - \tau_k} \tau_R \right) \frac{R_1^i}{1 + R_1^i}. \quad (\text{A49})$$

Thus, under this alternative base, a positive wealth tax lowers the reservation return and increases lock-in rather than undoing it.

Appendix E. Stochastic returns

This appendix presents a continuous-time version of our baseline model where returns are stochastic. This introduces two main changes to the setup. First, investors can hold their assets up to a time T so they face more than one realization choice. This makes it so that their problem not only depends on how their future return R_2 is relative to the market return r , but on how the future stream of returns will compare with realizing their gains up until t . This entails comparing realization today with the best possible realization strategy between t and T .

Second, returns are stochastic so that the problem of realization does not correspond to a deterministic choice, but to a stopping time rule. Therefore, lock-in and lock-out are no longer defined by comparing a reservation-return rule with the market rate. Instead, taxes distort the realization choice when the embedded capital gains of the asset (past returns) affect the realization decision at t , instead of that decision depending only on future returns.

For this, we define a new variable P_t that encodes the embedded capital gain of the asset. It corresponds to the cumulative returns (or the cumulative change in price) of the asset up to time t . Absent realization-based taxation, the embedded gain P is not payoff-relevant. Instead, the investor's problem retains the Markov structure of the canonical consumption–portfolio problem (Samuelson 1969; Merton 1969, 1971), with after-tax liquidation wealth and the private holding as states. Realization-based capital income taxation makes P_s be an additional state variable (Constantinides 1983; Dammon, Spatt, and Zhang 2001). This is the source of lock-in. We show removing capital income taxes ($\tau_k = 0$) or setting $\tau_a^* = (1 - \tau_k)r$ renders P redundant (Propositions A8 and A9 and Lemma A4). In these cases, taxes can still affect the dynamic consumption-savings choice of investors but there is no tax-based reason to make portfolio allocation depend on their embedded gains.

We verify our main results in this new setting, establishing that capital income taxes generate lock-in and that wealth taxes can undo it. We do this first in the context of maximizing expected returns by choosing the timing of capital gains, paralleling the analysis in the main text. We then extend the analysis to cover the portfolio choice of a risk-averse investor who can perform partial sales of their asset to manage risk. The same results hold under deterministic returns in discrete or continuous time.

E.1. Setup

Time is continuous on $[0, T]$. An investor is born with a private asset of value $A_0 > 0$ and can borrow or lend at the market return r . Realized capital gains are taxed once at rate $\tau_k \in [0, 1]$, while the historical value of wealth is taxed continuously at rate $\tau_a \geq 0$. The resulting after-tax return on the market asset is $\tilde{r} = (1 - \tau_k)r - \tau_a$. We set $\tau_R = 0$.

Returns. Let P_t denote the cumulative gross value of one dollar initially invested in the private asset, with $P_0 = 1$. In the deterministic case, it can be written as

$$P_t = \exp \left(\int_0^t R_u du \right), \quad P_0 = 1. \quad (\text{A50})$$

In the stochastic analysis below, P_t is instead defined directly by the diffusion in (A51). Thus, one dollar of private wealth held at date t has tax basis $1/p$ and embedded capital gain $1 - 1/p$. Here, p is the known value of P at time t . In our two-period setting the variable P corresponds to the term $1 + R_1$, so that the middle term in the reservation return in (8) corresponds to $1 - \frac{1}{p} = \frac{R_1}{1+R_1}$.

We consider a time-inhomogeneous geometric diffusion in which the expected private return is a known function of calendar time. This generalizes the standard geometric Brownian motion and allows us to nest unrestricted deterministic return paths when volatility is zero.

Specifically, let $\{W_s\}_{s \geq t}$ be a Brownian motion on a filtered probability space and let $\mu : [0, T] \rightarrow \mathbb{R}$ be bounded and piecewise continuous. The cumulative private value satisfies, conditional on information at date t ,

$$\frac{dP_s}{P_s} = \mu(s) ds + \sigma dW_s, \quad s \in [t, T], \quad (\text{A51})$$

with $P_t = p > 0$ and constant volatility $\sigma \geq 0$. Thus,

$$\frac{P_s}{p} = \exp \left[\int_t^s \left(\mu(u) - \frac{\sigma^2}{2} \right) du + \sigma(W_s - W_t) \right]. \quad (\text{A52})$$

When comparing realization incentives across values of p , we hold the stopping rule fixed as the same adapted functional of the normalized process $X_s \equiv P_s/p$. Since the law of X is independent of p , this comparison changes the embedded capital gain without changing the future-return process or the stopping strategy.

Finally, the standard geometric Brownian motion is the following special case, which we use to strengthen some of the results below.

Assumption A1. The expected private return is constant, $\mu(s) = \mu$ for every $s \in [0, T]$, so

$$\frac{dP_s}{P_s} = \mu ds + \sigma dW_s \quad \text{and} \quad \frac{P_s}{p} = \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) (s - t) + \sigma(W_s - W_t) \right]. \quad (\text{A53})$$

E.2. Portfolio efficiency

We start by tackling the problem of efficient portfolio choice that maximizes expected returns and hence expected resources. The linearity of resources on returns makes this an optimal stopping time problem. The stopping time is the random time $\theta \in [t, T]$,

adapted to the investor's information, at which the remaining private asset is realized. Terminal realization is imposed at T .

To isolate the realization margin, we value the marginal private position by its expected payoff in date- t liquid wealth units. The value of realizing one dollar of private wealth immediately is

$$S(P_t) = 1 - \tau_k \left(1 - \frac{1}{P_t} \right). \quad (\text{A54})$$

Waiting until time $\theta > t$ has an expected value

$$\mathcal{H}(t, p; \theta) = \mathbb{E}_{t,p} \left[e^{-\tilde{r}(\theta-t)} \left((1 - \tau_k) \frac{P_\theta}{p} + \frac{\tau_k}{p} \right) - \tau_a \int_t^\theta e^{-\tilde{r}(u-t)} \frac{P_u}{p} du \right]. \quad (\text{A55})$$

Here, $\mathbb{E}_{t,p}[\cdot]$ denotes expectations conditional on the investor's date- t information and on $P_t = p$. At realization, the current dollar has grown to P_θ/p and retains tax basis $1/p$, which gives the after-tax proceeds $(1 - \tau_k)P_\theta/p + \tau_k/p$. The integral is the discounted flow of historical wealth taxes paid while the private asset is retained. The stochastic optimal stopping problem is

$$\mathcal{V}(t, p) = \max \left\{ S(p), \sup_{\theta \in \mathcal{T}_t} \mathcal{H}(t, p; \theta) \right\}, \quad (\text{A56})$$

where $\mathcal{T}_t$ is the set of stopping times taking values in $(t, T]$.

For completeness, we state the variational inequality satisfied by the value per unit of original basis, $\mathcal{Y}(t, p) \equiv p\mathcal{V}(t, p)$ in the continuation region:

$$\max \left\{ \underbrace{(1 - \tau_k)p + \tau_k - \mathcal{Y}(t, p)}_{\text{Immediate Realization}}, \underbrace{\mathcal{Y}_t + \mu(t)p\mathcal{Y}_p + \frac{1}{2}\sigma^2 p^2 \mathcal{Y}_{pp} - \tilde{r}\mathcal{Y} - \tau_a p}_{\text{Holding the Asset}} \right\} = 0, \quad (\text{A57})$$

with terminal condition $\mathcal{Y}(T, p) = (1 - \tau_k)p + \tau_k$ for the forced realization at time T .

Realization choice: Capital lock-in and lock-out. For any stopping rule $\theta \in \mathcal{T}_t$, define the gain from continuation relative to immediate realization as

$$\mathcal{D}(t, p; \theta) \equiv \mathcal{H}(t, p; \theta) - S(p), \quad (\text{A58})$$

and define the optimized continuation wedge as

$$\mathcal{D}^*(t, p) \equiv \sup_{\theta \in \mathcal{T}_t} \mathcal{D}(t, p; \theta). \quad (\text{A59})$$

The investor weakly prefers immediate realization when $\mathcal{D}^*(t, p) \leq 0$ and strictly prefers retaining the private asset when $\mathcal{D}^*(t, p) > 0$.

We first characterize the no-tax realization rule. This serves as a useful point of comparison below, but it is not required for our general results.

Proposition A6 (No-tax realization rule). *Fix $t < T$ and set $\tau_k = \tau_a = 0$.*

Under (A51), the realization decision is independent of the embedded capital gain p . An optimal stopping date can be chosen deterministically to maximize the cumulative expected excess returns. The set of optimal dates is

$$\arg \max_{s \in [t, T]} \int_t^s (\mu(u) - r) du. \quad (\text{A60})$$

Immediate realization is optimal if and only if $\int_t^s (\mu(u) - r) du \leq 0$ for every $s \in (t, T]$.

Under Assumption A1, this result strengthens to the familiar three cases: if $\mu < r$, the investor realizes immediately; if $\mu > r$, the investor retains the private asset until T ; and if $\mu = r$, the investor is indifferent among all stopping times.

PROOF. Let $X_s \equiv P_s/p$ be the value of the private asset at date s relative to its value at date t , so that $X_t = 1$. By (A51), the law of X is independent of p . Without taxes, the continuation wedge is

$$\mathcal{D}^0(t, p; \theta) = \mathbb{E}_{t,p} \left[e^{-r(\theta-t)} X_\theta \right] - 1 = \mathbb{E}_{t,p} \left[\int_t^\theta e^{-r(u-t)} (\mu(u) - r) X_u du \right], \quad (\text{A61})$$

where we hold θ fixed as the same adapted functional of X by applying Itô's formula to $e^{-r(s-t)} X_s$ and using boundedness of $\theta \leq T$. The wedge, and therefore the stopping time, is independent of p because the distribution of (X, θ) does not depend on p .

We can provide a tighter characterization by defining

$$Z_s \equiv \exp \left[\sigma(W_s - W_t) - \frac{\sigma^2}{2}(s - t) \right]. \quad (\text{A62})$$

The process Z is a positive martingale with $Z_t = 1$. Equation (A52) implies

$$e^{-r(s-t)} X_s = e^{\int_t^s (\mu(u) - r) du} Z_s. \quad (\text{A63})$$

For any bounded stopping time $\theta \in \mathcal{T}_t$, optional sampling gives $\mathbb{E}_{t,p}[Z_\theta] = 1$. Therefore,

$$\mathbb{E}_{t,p} \left[e^{-r(\theta-t)} X_\theta \right] \leq \left(\max_{s \in [t, T]} e^{\int_t^s (\mu(u) - r) du} \right) \mathbb{E}_{t,p}[Z_\theta] = \max_{s \in [t, T]} e^{\int_t^s (\mu(u) - r) du}. \quad (\text{A64})$$

A deterministic date that maximizes $\int_t^s (\mu(u) - r) du$ attains this bound because $\mathbb{E}[Z_s] = 1$. This proves (A60) and also verifies that the value does not depend on p . Immediate realization is optimal exactly when the maximum is attained at $s = t$, where $\int_t^s (\mu(u) - r) du = 0$.

Under Assumption A1, $\int_t^s (\mu(u) - r) du = (\mu - r)(s - t)$. This expression is decreasing in s when $\mu < r$, increasing when $\mu > r$, and constant when $\mu = r$. In the last case, (A63) is a martingale, so every bounded stopping time has the same expected payoff. $\square$

We can now define lock-in and lock-out in this environment where future returns are independent of embedded gains.

Definition A1 (Lock-in and lock-out in continuous time). *For $p > 1$, a tax system generates an embedded-gain lock-in incentive if $\mathcal{D}(t, p; \theta)$ is weakly increasing in p for every fixed future stopping rule and strictly increasing for some relevant stopping rule. It generates an embedded-gain lock-out incentive if the inequalities are reversed.*

Relative to the no-tax allocation in Proposition A6, an investor is locked in at (t, p) if immediate realization is optimal without taxes but taxes make $\mathcal{D}^(t, p) > 0$. An investor is locked out if the no-tax investor strictly prefers continuation but taxes make immediate realization optimal.*

Taxation and realization choices. When $\tau_a = 0$ and $\tau_k > 0$, the continuation wedge for any stopping time $\theta > t$ is

$$\mathcal{H}(t, p; \theta) - S(p) = (1 - \tau_k) \left[\mathbb{E}_{t,p} \left[e^{-(1-\tau_k)r(\theta-t)} \frac{P_\theta}{p} \right] - 1 \right] + \frac{\tau_k}{p} \left[\mathbb{E}_{t,p} \left[e^{-(1-\tau_k)r(\theta-t)} \right] - 1 \right]. \quad (\text{A65})$$

The first term is the after-tax value of retaining the private asset. The second term is the discounted value of preserving the tax basis until realization. The dependence on p comes through the tax-basis term τ_k/p , or equivalently through the embedded tax liability $\tau_k(1 - 1/p)$ that is triggered by immediate realization. This is why there is capital lock-in: the decision to realize is no longer governed only by future returns, but also by the tax liability embedded in the asset.²⁰

Proposition A7 (Capital income taxes generate stochastic lock-in). *Fix $t < T$ and assume $0 < \tau_k < 1$, $\tau_a = 0$, $r > 0$, and $p > 1$. Under (A51), for every fixed $\theta \in \mathcal{T}_t$, the continuation wedge $\mathcal{D}(t, p; \theta)$ is strictly increasing in p . Capital income taxation therefore generates an embedded-gain lock-in incentive according to Definition A1.*

Suppose, in addition, that $\mu(\cdot)$ is right-continuous at t , immediate realization is optimal in the no-tax problem, and

$$\mu(t) > r - \tau_k r \left(1 - \frac{1}{p} \right) = rS(p). \quad (\text{A66})$$

Then the investor strictly prefers to delay realization under the capital income tax and is locked in relative to the no-tax allocation.

²⁰The first term in (A65) depends on the process for P_s/p . The result holds as long as this process is independent of embedded gains (i.e., past returns), as is the case for (A51). When future returns depend on past returns the realization choice naturally depends on them as well, regardless of tax incentives.

PROOF. Holding θ fixed as the same adapted functional of X , the first term in (A65) does not depend on p because the law of X is independent of p . Therefore,

$$\frac{\partial \mathcal{D}(t, p; \theta)}{\partial p} = \frac{\tau_k}{p^2} \left\{ 1 - \mathbb{E}_{t,p} \left[e^{-(1-\tau_k)r(\theta-t)} \right] \right\} > 0. \quad (\text{A67})$$

The inequality follows because $\theta > t$, $r > 0$, and $0 < \tau_k < 1$. Taking the supremum over θ preserves weak monotonicity in p .

To establish the local result, consider the deterministic stopping time $\theta = t + h$. Since $\mathbb{E}_{t,p}[X_{t+h}] = \exp(\int_t^{t+h} \mu(u) du)$, equation (A65) becomes

$$\mathcal{D}(t, p; t+h) = (1-\tau_k) \left[\exp \left(\int_t^{t+h} \mu(u) du - (1-\tau_k)rh \right) - 1 \right] + \frac{\tau_k}{p} \left[e^{-(1-\tau_k)rh} - 1 \right]. \quad (\text{A68})$$

Differentiating at $h = 0$ gives

$$\left. \frac{\partial \mathcal{D}(t, p; t+h)}{\partial h} \right|_{h=0} = (1-\tau_k) \left[\mu(t) - r + \tau_k r \left(1 - \frac{1}{p} \right) \right]. \quad (\text{A69})$$

Condition (A66) makes this derivative positive. A sufficiently short delay is therefore strictly preferable under the tax, while immediate realization is optimal without taxes by assumption. $\square$

The historical wealth tax undoes the embedded-gain incentive at every date just as before. Set $\tau_a = \tau_a^* = (1-\tau_k)r$, so that $\tilde{r} = 0$. For any stopping time $\theta \in \mathcal{T}_t$, subtracting the immediate-realization payoff from the continuation payoff gives

$$\begin{aligned} \mathcal{H}(t, p; \theta) - S(p) &= \mathbb{E}_{t,p} \left[(1-\tau_k) \frac{P_\theta}{p} + \frac{\tau_k}{p} - \int_t^\theta (1-\tau_k)r \frac{P_u}{p} du \right] - \left(1 - \tau_k + \frac{\tau_k}{p} \right) \\ &= (1-\tau_k) \mathbb{E}_{t,p} \left[\frac{P_\theta}{p} - 1 - r \int_t^\theta \frac{P_u}{p} du \right]. \end{aligned} \quad (\text{A70})$$

The term τ_k/p cancels exactly. Thus, the stopping rule no longer depends on the embedded capital gain $1 - 1/p$. The investor may still wait when the expected private return exceeds the market return, but the reason for waiting is no longer the capital-gains tax basis. So, the historical wealth tax removes the tax advantage of postponing realization.

Proposition A8 (Historical wealth taxes remove stochastic lock-in). *Fix $t < T$, assume $0 \leq \tau_k < 1$ and $r \geq 0$, and set $\tau_a^* = (1-\tau_k)r$. Under (A51), $\mathcal{D}(t, p; \theta)$ is independent of p for every stopping rule $\theta \in \mathcal{T}_t$. The historical wealth tax therefore eliminates embedded-gain lock-in and lock-out incentives at every date.*

Under Assumption A1, the investor's realization decision coincides with the no-tax realization decision.

PROOF. Let $X_s \equiv P_s/p$ be the value of the private asset at date s relative to its value at date t , so that $X_t = 1$. Equation (A70) is independent of p because the law of X is independent of p . Applying Itô's formula to X_s and using boundedness of $\theta \leq T$ gives

$$\mathcal{D}(t, p; \theta) = (1 - \tau_k) \mathbb{E}_{t,p} \left[\int_t^\theta (\mu(u) - r) X_u du \right]. \quad (\text{A71})$$

This proves embedded-gain neutrality for the general deterministic drift function.

Under Assumption A1, equation (A71) becomes

$$\mathcal{D}(t, p; \theta) = (1 - \tau_k)(\mu - r) \mathbb{E}_{t,p} \left[\int_t^\theta X_u du \right]. \quad (\text{A72})$$

Since $X_u > 0$, the sign of (A72) is determined entirely by $\mu - r$ as in Proposition A6. $\square$

The exact allocation equivalence in the second part of Proposition A8 is a special property of Assumption A1 and, more generally, of return paths for which $\mu(u) - r$ does not change sign over the relevant horizon. Without taxes, equation (A61) discounts each future excess return by $e^{-r(u-t)}$. At the unlocking wealth tax, equation (A71) does not contain this discount factor. When $\mu(u) - r$ changes sign, these two criteria may rank stopping dates differently. Thus, embedded-gain neutrality survives for an unrestricted deterministic drift path, but exact restoration of the no-tax stopping allocation depends on the path of returns for the investor's private asset.

E.3. Optimal portfolio choice

We consider the problem of a risk-averse investor and show that capital income taxes introduce an embedded gain lock-in effect that is negated by historical wealth taxes. This is different from the embedded gain effects exposed before that distort efficient portfolio choice that seeks to maximize expected returns. The problem is now to maximize an investor's expected utility discounted with $\delta > 0$

$$\mathbb{E}_t \left[\int_t^T e^{-\delta(s-t)} u(C_s) ds \right]. \quad (\text{A73})$$

The investor has liquid wealth B_s and a private asset with market value A_s . As before, let P_s in (A51) denote the cumulative value of one dollar initially invested in A , with $P_0 = 1$. Proportional sales of the asset A reduce the market value and historical basis by the same fraction so that the basis of the position remaining at date s is A_s/P_s . The embedded gain is therefore $A_s(1 - 1/P_s)$, or $1 - 1/P_s$ per dollar of current private wealth.

The investor chooses to realize (or liquidate) a fraction of its private asset at time s . In general, it is not optimal to liquidate the whole asset. For this, let L_s be a non-decreasing, right-continuous process, with $L_{t-} = 0$, that records proportional sales of the remaining private position. A sale dL_s reduces the market value and its associated historical basis

in the same proportion but leaves the cumulative-return process P_s unchanged. Before the private asset has been completely sold, the state dynamics are

$$dA_s = \mu(s)A_s ds + \sigma A_s dW_s - A_{s-} dL_s, \quad A_t = a > 0, \quad (\text{A74})$$

$$dB_s = [\tilde{r}B_s - \tau_a A_s - C_s] ds + \left[1 - \tau_k \left(1 - \frac{1}{P_s} \right) \right] A_{s-} dL_s, \quad B_t = b. \quad (\text{A75})$$

The same Brownian motion drives P_s and A_s . Applying Itô's formula therefore gives

$$d \left(\frac{A_s}{P_s} \right) = - \frac{A_{s-}}{P_s} dL_s. \quad (\text{A76})$$

Thus, A_s/P_s is constant between sales and falls only when part of the private position is liquidated. The last term in (A75) is the after-tax revenue from a sale captured by $S(P_s)$, the value of realization: the investor pays $\tau_k(1 - 1/P_s)A_{s-}dL_s$ on the realized gain $(1 - 1/P_s)A_{s-}dL_s$. We allow jumps satisfying $0 \leq \Delta L_s \leq 1$; a jump $\Delta L_s = 1$ liquidates the remaining position. Once $A_s = 0$, the sale process is constant. Admissible policies must nevertheless satisfy solvency, so that after-tax liquidation wealth, Y_s , is positive and the investor has resources to consume:

$$Y_s \equiv B_s + \left[1 - \tau_k \left(1 - \frac{1}{P_s} \right) \right] A_s = B_s + S(P_s)A_s > 0, \quad (\text{A77})$$

whenever resources remain, together with the usual integrability conditions.

The value of realization of gains, $S(p)$, is therefore crucial. It shapes the evolution of assets and is the way in which embedded gains P affect the investor's problem. This is more clearly seen by casting the investor's problem in terms of after-tax liquidation wealth, Y_s . Then the relevant states of the problem are (Y_s, A_s, P_s) with evolution for Y_s

$$dY_s = \left[\tilde{r}Y_s + A_s \left((1 - \tau_k)(\mu(s) - \tilde{r}) - \tau_a - \tilde{r} \frac{\tau_k}{P_s} \right) - C_s \right] ds + (1 - \tau_k) \sigma A_s dW_s \quad (\text{A78})$$

where the evolution of Y_s does not depend on the liquidation choice L_s because liquidating the private asset changes the composition of wealth but not its level.

Lock-in from capital income taxes. Critically, in the law of motion for after-tax liquidation wealth, the embedded gain P_s enters through the drift of wealth. This is more easily seen by expressing the middle term in the drift as

$$\left((1 - \tau_k)(\mu(s) - \tilde{r}) - \tau_a - \tilde{r} \frac{\tau_k}{P_s} \right) = (1 - \tau_k)\mu(s) - \tilde{r} - \tau_a + \tau_k \tilde{r} \left(1 - \frac{1}{P_s} \right). \quad (\text{A79})$$

When $P_s > 1$ and $\tau_a = 0$, this becomes the term $\tau_k r(1 - 1/P_s)$ in (A80), which raises the return from delaying realization without raising risk. This is the same term that distorted the efficient portfolio choice in (A58) and it would be absent without τ_k .

This is the *embedded-gain lock-in incentive*: a larger accrued gain makes realization less attractive because selling accelerates the capital income tax. This incentive is separate from risk and from the role of the capital income tax in sharing private-return risk between the government and the investor. Therefore, the embedded gain distorts the liquidation choice regardless of the investor's attitude toward risk. That is, two investors with the same Y_s and A_s , but different P_s want to make different choices.

Setting $\tau_k > 0$ and $\tau_a = 0$, so that $\tilde{r} = (1 - \tau_k)r$, equation (A78) simplifies to

$$dY_s = \left[(1 - \tau_k) r Y_s + (1 - \tau_k) A_s \left[\mu(s) - r + \tau_k r \left(1 - \frac{1}{P_s} \right) \right] - C_s \right] ds + (1 - \tau_k) \sigma A_s dW_s. \quad (\text{A80})$$

Unlocking capital with wealth taxes. Crucially, the *embedded-gain lock-in incentive* disappears when setting the historical wealth tax to $\tau_a^* = (1 - \tau_k)r$. Then (A78) reduces to

$$dY_s = \left[(1 - \tau_k) A_s (\mu(s) - r) - C_s \right] ds + (1 - \tau_k) \sigma A_s dW_s. \quad (\text{A81})$$

The wealth tax therefore removes the tax incentive to defer accrued gains and isolates the role of symmetric capital income taxes in risk sharing. This role of τ_k , along with the role of τ_a in changing the return on B_s , can still shape the allocation, so the complete solution of the investor's problem need not coincide with the no-tax allocation.

Portfolio choice without capital income taxes. Finally, setting $\tau_k = 0$, (A78) becomes

$$dY_s = \left[(r - \tau_a) Y_s + A_s (\mu(s) - r) - C_s \right] ds + \sigma A_s dW_s. \quad (\text{A82})$$

Just as in the main paper, τ_a does not distort the realization choice and only affects the intertemporal return through the drift of wealth.

Proposition A9 (Embedded-gain independence). *Suppose $0 < \tau_k < 1$.*

When $\tau_a = 0$, there is an incentive to defer embedded gains because an appreciated position receives the additional return $\tau_k r (1 - 1/P_s)$ in (A80).

When $\tau_a = \tau_a^$, the optimal consumption and realization policies can be written as functions of s , A_s , and the after-tax liquidation value Y_s , without depending separately on P_s . Thus, the wealth tax eliminates the embedded-gain deferral incentive and makes the embedded gain irrelevant for the investor's problem after accounting for current after-tax wealth. The value function and an optimal policy depend only on s , A_s , and Y_s .*

PROOF. The dynamics in (A80) follow by substituting $\tau_a = 0$ and $\tilde{r} = (1 - \tau_k)r$ into (A78). Holding A_s and Y_s fixed, an increase in P_s raises the drift from retaining the private position through $\tau_k r (1 - 1/P_s)$ without changing its instantaneous volatility.

At $\tau_a = \tau_a^*$, the after-tax liquidation-value dynamics are given by (A81) and contain no separate dependence on P_s . The existing law of motion for A_s in (A74) is also independent of P_s . $\square$

E.4. Investor's problem

For completeness, we now set up the investor's problem in terms of its Hamilton-Jacobi-Bellman representation and the corresponding variational inequality.

The liquid-wealth problem. Before characterizing the investors' portfolio problem, we solve the simpler problem of an investor who has sold all of their private asset. Let $J(t, b)$ denote the value of liquid wealth b in that case. This will serve as a boundary condition for the portfolio problem below. The investor solves

$$J(t, b) = \sup_{\{C_s\}_{s \in [t, T]}} \int_t^T e^{-\delta(s-t)} u(C_s) ds, \quad \text{s.t. } dB_s = (\tilde{r}B_s - C_s)ds, \quad B_t = b > 0. \quad (\text{A83})$$

The boundary condition is $J(T, b) = 0$ because there is no utility from terminal wealth. This problem has a closed form solution under standard CRRA utility.

Investor's consumption and liquidation problem. Let $V(t, y, a, p)$ be the value before A has been completely sold, where y denotes after-tax liquidation wealth as defined in (A77). The investor chooses consumption and the cumulative sale process:

$$V(t, y, a, p) = \sup_{\{C_s, L_s\}} \mathbb{E}_t \left[\int_t^T e^{-\delta(s-t)} u(C_s) ds \right], \quad (\text{A84})$$

subject to (A51), (A74), (A78), and the admissibility condition $Y_s > 0$. The value of a marginal proportional sale is

$$\mathcal{G}V(t, y, a, p) \equiv -aV_a(t, y, a, p). \quad (\text{A85})$$

Because y is after-tax liquidation wealth, a proportional sale changes the composition of wealth but not y itself. It only reduces the remaining private exposure a and leaves p unchanged. Let the drift of Y be denoted by

$$m(t, y, a, p) \equiv \tilde{r}y + a \left((1 - \tau_k)(\mu(t) - \tilde{r}) - \tau_a - \tilde{r} \frac{\tau_k}{p} \right). \quad (\text{A86})$$

The value function satisfies the gradient-constraint variational inequality

$$\begin{aligned} 0 = \max \left\{ \mathcal{G}V, V_t - \delta V + \sup_{C > 0} [u(C) - CV_y] + m(t, y, a, p)V_y + \mu(t)(aV_a + pV_p) \right. \\ \left. + \frac{1}{2} \sigma^2 \left[(1 - \tau_k)^2 a^2 V_{yy} + a^2 V_{aa} + p^2 V_{pp} + 2(1 - \tau_k)a^2 V_{ya} + 2(1 - \tau_k)apV_{yp} + 2apV_{ap} \right] \right\} \end{aligned} \quad (\text{A87})$$

The cross-derivative terms appear because Y_s , A_s , and P_s are all exposed to the same Brownian shock rather than to independent risks.

The first expression inside the maximum, $\mathcal{G}V$, is the marginal change in value from increasing the sale process L_s , including through a block sale that liquidates part of the private asset. The second expression is the Hamilton–Jacobi–Bellman equation when the sale process does not change: it determines the value while the continuous states evolve and consumption is chosen optimally.

In the no-sale region, $\mathcal{G}V < 0$ and the second line of (A87) equals zero. Sales occur when $\mathcal{G}V = 0$. The boundary value after complete liquidation is $V(t, y, 0, p) = J(t, y)$.

Lemma A4 (Redundant embedded-gain state). *If either $\tau_k = 0$ or $\tau_a = \tau_a^* = (1 - \tau_k)r$, the investor’s continuation problem can be written in terms of (t, Y_s, A_s) alone. In these cases, P_s does not enter the law of motion for Y_s , the law of motion for A_s , the sale condition, the solvency condition, or the boundary value after complete liquidation.*

PROOF. If $\tau_k = 0$, then $S(P_s) = 1$, so $Y_s = B_s + A_s$ and

$$dY_s = [(r - \tau_a)Y_s + A_s(\mu(s) - r) - C_s] ds + \sigma A_s dW_s. \quad (\text{A88})$$

This law of motion contains no P_s . If instead $\tau_a = \tau_a^* = (1 - \tau_k)r$, then $\tilde{r} = 0$ and (A78) reduces to

$$dY_s = [(1 - \tau_k)A_s(\mu(s) - r) - C_s] ds + (1 - \tau_k)\sigma A_s dW_s, \quad (\text{A89})$$

which also contains no P_s . In both cases, the law of motion for A_s in (A74) depends only on A_s and the sale process, a sale changes only A_s when the problem is written in terms of Y_s , and the boundary value is $J(t, Y_s)$. Therefore P_s is a redundant state variable. $\square$

The reduced value function $\bar{V}(t, y, a)$ solves

$$0 = \max \left\{ \mathcal{G}\bar{V}, \bar{V}_t + m(t, y, a)\bar{V}_y + \mu(t)a\bar{V}_a + \frac{1}{2}a^2 \left[\nu^2\bar{V}_{yy} + \sigma^2\bar{V}_{aa} + 2\nu\sigma\bar{V}_{ya} \right] - \delta\bar{V} + \sup_{C>0} [u(C) - C\bar{V}_y] \right\}, \quad (\text{A90})$$

with boundary value $\bar{V}(t, y, 0) = J(t, y)$. When $\tau_k = 0$,

$$m(t, y, a) = (r - \tau_a)y + a(\mu(t) - r), \quad \nu = \sigma. \quad (\text{A91})$$

When $\tau_a = \tau_a^*$,

$$m(t, y, a) = (1 - \tau_k)a(\mu(t) - r), \quad \nu = (1 - \tau_k)\sigma. \quad (\text{A92})$$

With no utility from terminal wealth, the terminal boundary is $\bar{V}(T, y, a) = 0$.